Partial Solutions to UCLA Basic Exam

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§1 Fall 2014

Problem 1.1. *Show that the function*

$$H(x,y) = x^2 + y^2 + \frac{1}{|x-y|}$$

achieves its global minimum somewhere on the set $S = \{x \in \mathbb{R}^2 : x \neq y\}$.

Proof. H attains a local minimum on the region $X \subset S$ where $1 \geq |x-y| \geq \frac{1}{2}$, $|y| \leq 1$ and $|x| \leq 1$ by the extreme value theorem. It suffices to show that the global minimum is not achieved in the complement of X. In the region where $|x| > 1$ and $|y| > 1$ we have $H(x,y) > 2$. In the region $|x-y| < \frac{1}{2}$ we again have $H(x,y) > 2$. On the other hand $(\frac{1}{2}, -\frac{1}{2}) \in X$ and $H(\frac{1}{2}, -\frac{1}{2}) = \frac{3}{2}$ (this is in fact the global minimum, but we merely need existence). □

Problem 1.2. *Let* A, B *be two closed subsets of* $\mathbb{R}^n$ *such that* $A \cup B$ *and* $A \cap B$ *are connected. Prove that* A *is connected.*

Proof. Assume $A \cap B \neq \emptyset$, for otherwise the claim is trivial. Suppose A is disconnected and $A = U \sqcup V$ is a separation of A for U, V nonempty closed subsets. Since $A \cap B = (U \cap B) \sqcup (V \cap B)$, either $U \cap B$ or $V \cap B$ is empty, say $U \cap B = \emptyset$. U and V are closed in $A \cup B$ because $A \cup B$ is closed in $\mathbb{R}^n$. It follows that $A \cup B = (U \cup B) \sqcup V$ is a separation of $A \cup B$, a contradiction. Hence A is connected. □

Problem 1.3. *Let* $f_n : [0,1] \to \mathbb{R}$ *be monotonically increasing continuous functions. Assume that* f_n *converge pointwise to a continuous function* $f : [0,1] \to \mathbb{R}$. *Prove that* f_n *converge uniformly to* f.

Proof. Since f is continuous on a compact set, it is uniformly continuous. Let $\varepsilon > 0$. Choose a partition $0 = a_1 < \cdots < a_k = 1$ such that in each $i = 1, 2, \ldots, k-1$ we have $|f(x) - f(y)| < \varepsilon$ for all $x, y \in [a_i, a_{i+1}]$. By pointwise convergence there exists an N such that $|f_n(a_i) - f(a_i)| < \varepsilon$ for each $i = 1, 2, \ldots, k$ and for all $n > N$. Since f_n is a monotone increasing function, if $x \in [a_i, a_{i+1}]$ we have

$$f_n(a_i) \leq f_n(x) \leq f_n(a_{i+1}).$$

$f_n(a_i)$ (resp. $f_n(a_{i+1})$ is within ε of $f(a_i)$ (resp. $f(a_{i+1})$), which is within ε of $f(x)$, so it follows that $|f_n(x) - f(x)| < 2\varepsilon$ for all $n > N$. Since ε was arbitrary, f_n converges uniformly to f.

□

Problem 1.4. *Let* $f_n : [-2,2] \to [0,1]$ *be a sequence of convex functions. Show that there is a subsequence which converges uniformly on* $[-1,1]$.

Proof. First let us see that f_n is (Lipschitz) continuous on $[-1,1]$. Convexity means the line segment between any two points on the graph of f_n lies above or on the graph. An equivalent condition is that the function

$$\varphi(x,y) = \frac{f_n(y) - f_n(x)}{y - x}$$

is non-decreasing as a function of y when x is fixed. Note that $\varphi(x,y) = \varphi(y,x)$. Thus if $y' > x', y > x, x' \geq x$, and $y' \geq y$. then

$$\frac{f_n(y) - f_n(x)}{y - x} = \varphi(x,y) \leq \varphi(x,y') \leq \varphi(x',y') = \frac{f_n(y') - f_n(x')}{y' - x'}.$$

We then have the bound $\varphi(-2,-\frac{3}{2}) \leq \varphi(x,y) \leq \varphi(\frac{3}{2},2)$ for all $x,y \in [-1,1]$, which shows that $|\varphi(x,y)|$ is bounded by $M = \max\{\varphi(\frac{3}{2},2), \varphi(-2,-\frac{3}{2})\}$. It follows that f_n is Lipschitz on $[-1,1]$. From this, it is obvious that the f_n are equicontinuous on $[-1,1]$. Since pointwise-boundedness is immediate, the result follows from Arzelà-Ascoli.

□

Remark 1.5. *That a convex function is Lipschitz is geometrically obvious from the double-cone condition for Lipschitz continuity, since convexity means the graph of the function lies between a horizontal double-cone with its vertex on the graph.*

Problem 1.6. *Consider the following sequence:*

$$a_1 = \sqrt{2} \quad \text{and} \quad a_{n+1} = \sqrt{2 + a_n} \quad \text{for all } n \geq 1.$$

Prove that this sequence converges and find its limit.

Proof. Assuming the limit L exists then letting $n \to \infty$ we get $L^2 = 2 + L$, so discarding the negative root $L = 2$.
For existence, we prove a_n is monotone increasing and bounded above by induction. Clearly $a_1 \leq 2$ and $a_1 \leq a_2$. Then by induction $a_n = \sqrt{2 + a_{n-1}} \leq \sqrt{2 + a_n} = a_{n+1}$ and $a_{n+1} = \sqrt{2 + a_n} \leq \sqrt{2+2} = 2$. Thus a_n converges, and must have limit 2. □

Problem 1.7. *Let $f : [0,1] \to \mathbb{R}$ be $\mathcal{C}^1$. Prove that*

$$\lim_{n\to\infty} \sum_{k=0}^{n-1} \left| f\left(\frac{k+1}{n}\right) - f\left(\frac{k}{n}\right) \right| = \int_0^1 |f'(x)|\, dx.$$

Proof. Let P_n be the partition $0 < \frac{1}{n} < \frac{2}{n} < \cdots < \frac{n}{n} = 1$. For each $k = 0, 1, \ldots, n-1$ there is a $\xi_k \in [\frac{k}{n}, \frac{k+1}{n}]$ such that

$$\frac{1}{n} \cdot f'(\xi_k) = f\left(\frac{k+1}{n}\right) - f\left(\frac{k}{n}\right)$$

by the mean value theorem. Then the corresponding Riemann sum for the RHS is

$$\sum_{k=0}^{n-1} \frac{1}{n} \cdot |f'(\xi_k)| = \sum_{k=0}^{n-1} \left| f\left(\frac{k+1}{n}\right) - f\left(\frac{k}{n}\right) \right|.$$

Since $|f'|$ is certainly Riemann integrable, this sum must converge to the integral as $n \to \infty$ and the mesh of P_n goes to 0. □

Problem 1.8. *Among all solutions $x \in \mathbb{R}^4$ to*

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 7 \\ -2 & -1 & 1 & 3 \end{pmatrix} x = \begin{pmatrix} 2 \\ 7 \\ -1 \end{pmatrix}$$

find the solution with minimum (Euclidean) length.

Proof. This is just (horribly boring) optimization. The nullspace of the LHS 3×4 matrix is spanned by the two vectors

$$v_1 = \begin{pmatrix} 2 \\ -3 \\ 1 \\ 0, \end{pmatrix} \qquad v_2 = \begin{pmatrix} 4 \\ -5 \\ 0 \\ 1 \end{pmatrix}.$$

A particular solution is $v = (0, 0, \frac{3}{2}, \frac{1}{2})$. Thus the general solution is

$$x(s,t) = v + tv_1 + sv_2$$

for constants t, s. We now minimize $N(s,t) := ||x(s,t)||$

$$\begin{aligned} N(s,t) &= (2t + 4Cs)^2 + (-3t - 5s)^2 + \left(t + \frac{3}{2}\right)^2 + \left(\frac{1}{2} + s\right)^2 \\ &= 14t^2 + \frac{5}{2} + s + 42s^2 + t(3 + 46s). \end{aligned}$$

Solving $\partial_s N = 0$ and $\partial_t N = 0$ we obtain the minimum at $(t, s) = (-103/118, 55/118)$. □

Problem 1.9. *Compute the eigenvalues of the $n \times n$ matrix*

$$M = \begin{pmatrix} k & 1 & 1\dots & 1 \\ 1 & k & 1\dots & 1 \\ 1 & 1 & k\dots & 1 \\ \vdots & \vdots & \ddots & 1 \\ 1 & 1 & 1\dots & k \end{pmatrix}$$

and compute $\det(M)$.

Proof. The eigenvectors of M are easily seen to be

$$(-1, 1, 0, \dots 0), (-1, 0, 1, 0, \dots, 0), (-1, 0, 0, 1, 0 \dots, 0), \dots, (1, 1, 1, \dots, 1).$$

with respective eigenvalues $k - 1$ with multiplicity $n - 1$ and $k + (n - 1)$ with multiplicity 1. Thus the determinant is given by $\det M = (k - 1)^{n-1}(k + n - 1)$. □

Problem 1.10. *Let $A \neq 0$ be an $n \times n$ complex matrix. Prove that there is a matrix B, such that B and $A + B$ have no eigenvalues in common.*

Proof. We begin by establishing the following claim: For every matrix having form

$$0 \neq M = \begin{pmatrix} 0 & 1 & 0\dots & 0 \\ 0 & 0 & 1\dots & 0 \\ 0 & 0 & 0\dots & 0 \\ \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0\dots & 0 \end{pmatrix}$$

that is, M has the form of a Jordan block corresponding to 0 as eigenvalue, there is some N such that $M + N$ and N have no common eigenvalue. To see this, consider

$$N = \begin{pmatrix} 0 & -1 & 0\dots & 0 \\ 0 & 0 & -1\dots & 0 \\ 0 & 0 & 0\dots & 0 \\ \vdots & \vdots & \ddots & -1 \\ -1 & 0 & 0\dots & 0 \end{pmatrix}$$

Then N is invertible and thus 0 is not an eigenvalue of N. However,

$$M + N = \begin{pmatrix} 0 & 0 & 0\dots & 0 \\ 0 & 0 & 0\dots & 0 \\ 0 & 0 & 0\dots & 0 \\ \vdots & \vdots & \ddots & 0 \\ -1 & 0 & 0\dots & 0 \end{pmatrix}$$

having only 0 as eigenvalue. Thus we establish the claim. Now for an arbitrary matrix A, consider the Jordan decomposition $A = SJS^{-1}$, where J is the Jordan matrix. If A is invertible, then 0 is not an eigenvalue of A, and $A + I = S(J + I)S^{-1}$. Notice that $J + I$ is an upper triangular matrix with no 1 on the diagonal, so 1 is not an eigenvalue of $A + I$, but 1 is the only eigenvalue of I. Therefore taking $B = I$ suffices. If A is not invertible, A contains Jordan blocks of the above form in the claim. We perturb the other Jordan blocks by the identity matrix, and the blocks corresponding to 0 with matrices having form of N above. Thereby we obtain a matrix K. It can be easily verified that $J + K$ and J have no common eigenvalue. Define $B = SKS^{-1}$ will give us the desired result. If the Jordan block corresponding to 0 only has size 1, (to be finished) □

Problem 1.11. *What is the largest number of* 1*s an invertible* 0-1 *matrix of size* $n \times n$ *can have? You must show both that this number is possible and that no larger number is possible.*

Proof. The number of 1's cannot exceed $n^2 - (n - 1)$, in which case two columns are all 1's. This maximum is attained by the following matrix:

$$M = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 0 & 1 & \dots & 1 \\ 1 & 1 & 0 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & 0 \end{pmatrix}$$

with 0's on the diagonal except the first entry. To see the matrix is non-singular, denote the first column vector by v. Then $M = (v, v - e_2, v - e_3, \dots, v - e_n)$ where e_i are the standard basis vectors. The columns are linearly independent, hence $\det M \neq 0$. □

Problem 1.12. *Suppose a* 4×4 *integer matrix has four distinct real eigenvalues* $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$. *Prove that* $\lambda_1^2 + \lambda_2^2 + \lambda_3^2 + \lambda_4^2$ *is an integer.*

Proof. Denote the matrix by M. Clearly the characteristic polynomial $p(x) = \det(M - xI)$ is a polynomial with integer coefficients . On the other hand, by Vietá's formulas, $a_3 = -(\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4)$. and $a_2 = \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_4 + \lambda_4\lambda_1$, where a_n is the coefficient of x^n. Thus

$$\lambda_1^2 + \lambda_2^2 + \lambda_3^2 + \lambda_4^2 = a_3^2 - 2a_2$$

is an integer. □

Problem 1.13. *Let* $A = a_{ij},\ 1 \leq i, j \leq n$, *where* $a_{ij} = 1/(i + j - 1)$. *Prove that* A *is positive definite.*

Proof. The matrix is

$$A = \begin{pmatrix} 1 & \frac{1}{2} & \cdots & \frac{1}{n} \\ \frac{1}{2} & \frac{1}{3} & \cdots & \frac{1}{n+1} \\ \vdots & \vdots & \ddots & \ddots \\ \frac{1}{n} & \frac{1}{n+1} & \cdots & \frac{1}{2n-1}. \end{pmatrix}$$

Then for $v = (v^1, \ldots, v^n) \neq 0$ we have

$$\begin{aligned} v^T A v &= \sum_{1 \leq i,j \leq n} \frac{v^i v^j}{i+j-1} = \sum_{1 \leq i,j \leq n} v^i v^j \int_0^1 x^{i+j-2}\, dx \\ &= \int_0^1 \left(\sum_{i=1}^n v^i x^{i-1} \right)^2 dx > 0 \end{aligned}$$

□

§2 Spring 2014

Problem 2.1. *(a) Find a real matrix A whose minimal polynomial is equal to $t^4 + 1$.*

(b) Show that the usual real linear map determined by $v \mapsto Av$ has no non-trivial invariant subspace.

Proof. 1. We can take a rotation counter-clockwise by $\pi/8$ in $\mathbb{R}^2$, given by

$$A = \begin{pmatrix} \cos(\pi/8) & -\sin(\pi/8) \\ \sin(\pi/8) & \cos(\pi/8) \end{pmatrix}.$$

2. Suppose there is a non-trivial invariant subspace V, then it is easy to see that $\dim V = 1$. Then there is a vector v such that $V = \mathrm{Span}\{v\}$. If we want V to be an invariant subspace, we must have $Av = \lambda v$ for some $\lambda \in \mathbb{R}$, but this is impossible since A has no real eigenvalue.

□

Problem 2.2. *Suppose that $S, T \in Hom(V, V)$ where V is a finite dimensional vector space over $\mathbb{R}$. Show that*

$$\dim(\mathrm{Im}\, S) + \dim(\mathrm{Im}\, T) \leq \dim(\mathrm{Im}(S \circ T)) + \dim V$$

Proof. This is Sylvester ranking inequality. Instead of directly showing the result in the problem we show equivalently

$$\dim \ker S \circ T \leq \dim \ker S + \dim \ker T$$

To show this, suppose $\{t_1, t_2, ..., t_r\}$ is a basis for $\ker T$. Since $\ker T \subset \ker S \circ T$, we may extend the above basis to $\{t_1, ..., t_r, u_1, ..., u_s\}$ a basis of $\ker S \circ T$. Note that $T(u_i) \in \ker S$ for all $1 \leq i \leq s$, we claim that $\{T(u_1), ..., T(u_s)\}$ is linearly independent. To show the claim we suppose $a_1 T(u_1) + ... + a_s T(u_s) = 0$, then $T(a_1 u_1 + ... + a_s u_s) = 0$, meaning that $a_1 u_1 + ... + a_s u_s \in \ker T$. However, $t_1, ..., t_r, u_1, ..., u_s$ are linearly independent in V, so $a_i = 0$ for all i. Then we show the claim. Substitution gives

$$\dim \ker S \circ T - \dim \ker T \leq \dim \ker S$$

and moving the terms we get the desired result.

□

Problem 2.3. *Suppose $A, B \in M_{n\times n}(\mathbb{C})$ satisfies $AB - BA = A$. Show that A is not invertible.*

Proof. Suppose on the contrary that A is invertible, then

$$B = A^{-1}(I + B)A$$

and since trace is invariant under similarity we have

$$\operatorname{trace} B = \operatorname{trace}(I + B)$$

which is impossible. Thus A cannot be invertible, as desired. □

Problem 2.4. *Suppose $A, B \in M_{n\times n}(\mathbb{C})$. Show that the characteristic polynomials of AB and BA are equal.*

Proof. First of all suppose B is invertible, then $\det(\lambda I - AB) = 0$ if and only if $\det(\lambda B^{-1}A) = 0$ if and only if $\det(\lambda I - BA) = 0$. This means that AB and BA have the same eigenvalues. If B is not invertible, note that $GL(n, \mathbb{R})$ is dense in $M_{n\times n}(\mathbb{C})$, where the topology is entry-wise convergence. We take a sequence $\{B_n\}_n$ converging to B, and passing the above result to limit we get the desired conclusion. □

Problem 2.5. *Let V be a real inner product space and let $L \in Hom(V, V)$. Fix $b \in V$. Prove that if $u, v \in V$ both minimize $D(x) = ||L(x) - b||$ then $u - v \in \ker L$.*

Proof. For convenience we denote $U := L(V)$. We claim that $\|L(x) - b\|$ is minimized if and only if $L(x) = P_U(b)$, where $P_U(b)$ denotes the orthogonal projection onto U. To show the claim note that

$$\begin{aligned}\|b - P_U(b)\|^2 &\leq \|b - P_U(b)\|^2 + \|P_U(b) - L(x)\|^2 \\ &= \|b - P_U(b) + P_U(b) - L(x)\|^2 \\ &= \|b - L(x)\|^2\end{aligned}$$

where the first equality is by the fact that $b - P_U(b) \in U^\perp$ and $P_U(b) - L(x)$ and applying Pythogorean theorem. Since the equality holds if and only if $\|P_U(b) - L(x)\| = 0$ if and only if $P_U(b) = L(x)$, we prove the claim. By the claim we have that $\|L(x) - b\|$ is minimized if and only if $L(x) = P_U(b)$, and thus if u and v both minimize this quantity, $L(u) = L(v)$ and hence $u - v \in \ker L$. This finishes the proof. □

Problem 2.6. *Show that if $A \in M_{n\times n}(\mathbb{C})$ is normal, then $A^* = P(A)$ for some $P \in \mathbb{C}[x]$, where A^* is the conjugate transpose of A.*

Proof. Since $A \in M_n(\mathbb{C})$ is normal, by complex spectral theorem and Schur decomposition there exists unitary matrix U such that $A = UDU^{-1}$ where D is diagonal. Then note that

$$A^* = (UDU^{-1})^* = UD^*U^*$$

Suppose A has eigenvalue $\lambda_1, ..., \lambda_k$, consider a polynomial P such that $P(\lambda_i) = \overline{\lambda_i}$ for all i, whose existence is granted by Lagrange interpolation formula. Then

$$P(A) = UP(D)U^{-1} = UD^*U^{-1} = UD^*U^* = A^*$$

as desired. □

Problem 2.7. *Find a doubly infinite sequence $\{a_{n,m}\}_{n,m\in\mathbb{Z}}$ such that for all m, $\sum_n a_{n,m} = 0$ and for all n, $\sum_m a_{n,m} = 0$ where these series absolutely converge, but such that*

$$\sum_n \sum_m |a_{n,m}| = \infty$$

Proof. Define a_{nm} as follows:

$$a_{nm} = \begin{cases} 1 & n = m \\ -1 & n = m-1 \\ 0 & \text{otherwise} \end{cases}$$

Then the sum over m for fixed n, and the sum over n for fixed m is clearly 0. However, $\sum_{n,m} |a_{n.m}|$ diverges. □

Problem 2.8. *(a) Define $f : \mathbb{R} \to \mathbb{R}$ by*

$$f(x) = \begin{cases} 0, & x \le 0 \\ e^{-1/x}, & x > 0 \end{cases}.$$

Prove that f is infinitely differentiable.

(b) In Euclidean space $\mathbb{R}^n$, find a function $\varphi \in \mathcal{C}^\infty$ such that $\varphi(x) = 0$ if $|x| > 1$ and

$$\int_{\mathbb{R}^n} \varphi(x)\, dx = 1.$$

Proof. □

For (a) it suffices to check f is smooth at $x = 0$. By L'Hopital and induction,

$$\lim_{x\to 0^+} \frac{e^{-1/x}}{x^k} = 0$$

for any non-negative integer k. In particular f is continuous at 0. We claim that for $x > 0$ the k-th derivative of f is of the form

$$f^{(k)}(x) = P_k(x)\frac{e^{-1/x}}{x^{2k}}$$

for some polynomial P_k with $\deg P_k \le k$. $k = 0$ is trivial. Moreover

$$\begin{aligned} f^{(k+1)}(x) &= P_k'(x)\frac{e^{-1/x}}{x^{2k}} + P_k(x)\frac{x^{-2}e^{-1/x}}{x^{2k}} - 2kP_k(x)\frac{e^{-1/x}}{x^{2k+1}} \\ &= (x^2P_k'(x) + P_k(x) - 2kxP_k(x))\frac{e^{-1/x}}{x^{2(k+1)}} \end{aligned}$$

so the claim follows by induction. Now note that the right-sided derivative $f^{(k+1)}$ is given by

$$\lim_{x\to 0^+} \frac{1}{x}\frac{e^{-1/x}}{x^{2k}} = P_k(0) \lim_{x\to 0^+} \frac{e^{-1/x}}{x^{2k+1}}.$$

The RHS is the right-sided derivative $f^{(k)}(0)$. Since $f^{(0)}(0) = 0$ it follows by induction that the right-sided derivatives of all order are 0 at $x = 0$. The left-sided limit is obviously 0, so we conclude

that $f^{(k)}(0) = 0$ for all k. Hence f is smooth.

For (b) define

$$g(x) = \frac{f\left(\frac{1}{2} - x\right)}{f\left(\frac{1}{2} - x\right) + f\left(x - 1\right)}.$$

Clearly g is bounded and $g(x) = 0$ for all $|x| > 1$. Moreover $I = \int_{\mathbb{R}^n} g$ is finite and positive. Letting $\varphi = g/I$ gives the result.

Problem 2.9. *Find a function $f \in \mathcal{C}^1$ with $f(0) = 0$ and $f(1) = 1$ that minimizes*

$$\int_0^1 |f'(x)|^2 \, dx.$$

Is the minimizing function unique?

Proof. By Cauchy-Schwarz inequality

$$\int_0^1 |f'(x)|^2 \, dx \geq \left(\int_0^1 |f'(x)|^2 \, dx\right)\left(\int_0^1 1^2 \, dx\right) \geq \left(\int_0^1 f'(x) \, dx\right)^2 = 1$$

and according to the equality condition of Cauchy-Schwartz, the minimum is achieved when $f'(x) = \lambda$ for $\lambda \in \mathbb{R}$. Since $f(0) = 0$ and $f(1) = 1$, $f'(x) = 1$ and $f(x) = x$. Since the equality condition is sufficient and necessary, the miminizing function is unique. □

Problem 2.10. *Let $\mathcal{F} = \mathcal{C}([0, 1])$. Assume that*

(i) $\mathcal{F}$ is uniformly bounded.

(ii) $\mathcal{F}$ is equicontinuous.

Prove that every sequence in $\mathcal{F}$ has a subsequence that converges uniformly on $[0, 1]$.

Proof. (This is Arzelà-Ascoli). Let $\{f_n\}$ be a sequence of functions in $\mathcal{F}$ and let $E = \mathbb{Q} \cap [0, 1]$. By a diagonalization argument it's easy to see there is a subsequence $\{f_{n_i}\}$ such that $\{f_{n_i}(x)\}$ converges for all $x \in E$. Let $\varepsilon > 0$ and choose $\delta > 0$ such that $|f_n(x) - f_n(y)| < \varepsilon$ whenever $|x - y| < \delta$, which is possible by equicontinuity. Since E is dense we can choose $q_1, \ldots, q_k \in E$ such that

$$[0, 1] \subset B(q_1, \delta) \cup \ldots B(q_k, \delta).$$

Choose N large enough so that $|f_{n_i}(q_m) - f_{n_j}(q_m)| < \varepsilon$ for all $n_i > N$ and $m = 1, \ldots, k$.

Now let $x \in [0, 1]$ be arbitrary. There is some q_m such that $|x - q_m| < \delta$. Thus

$$|f_{n_i}(x) - f_{n_j}(x)| \leq |f_{n_i}(x) - f_{n_i}(q_m)| + |f_{n_i}(q_m) - f_{n_j}(q_m)| + |f_{n_j}(q_m) - f_{n_j}(x)| < 3\varepsilon.$$

Therefore the subsequence converges uniformly by the Cauchy criterion. □

Problem 2.11. *Prove the converse of Problem* 2.10.

Proof. To show uniform boundedness, suppose in the contrary that $\mathcal{F}$ is not uniformly bounded. For each $n \in \mathbb{N}$, pick $f_n \in \mathcal{F}$ such that $\|f_n\| \geq n$, where the norm here is the uniform norm. For each n, between n and $n + 1$ there can only be finitely many elements in the sequence. Then every subsequence of $\{f_n\}_n$ must eventually blow up to ∞ and hence cannot be convergent.

We then show that $\mathcal{F}$ is equicontinuous. We begin with the following claim:
Claim. $\overline{\mathcal{F}}$ is compact.

Proof of Claim. Since we already know sequences in $\mathcal{F}$ have uniformly convergent subsequence, to use Bolzano-Weierstrass it suffices to check sequences such that all but finitely elements are on $\partial\mathcal{F}$. Let $\{f_n\}_n$ be such a sequence. We wish to show that $\{f_n\}_n$ contains a uniformly convergent subsequence. For each f_n, take $B(f_n, 1/n)$, and since $f_n \in \overline{\mathcal{F}}$, there exists $f'_n \in B(f_n, 1/n) \cap \mathcal{F}$. Being in $\mathcal{F}$, $\{f'_n\}_n$ contains a uniformly convergence subsequence which we denote by $\{f'_{n_k}\}_k$. We shall show that $\{f_{n_k}\}_k$ is a uniformly convergent subsequence. Let $\epsilon > 0$. There exists $N \in \mathbb{N}$ such that when $k, l > N$, $\|f'_{n_k} - f'_{n_l}\|_u < \epsilon$. Take $N' \in \mathbb{N}$ large enough that $N' \geq N$ and $\frac{1}{N'} < \epsilon$, for $k, l > N'$,

$$\|f_{n_k} - f_{n_l}\|_u \leq \|f_{n_k} - f'_{n_k}\|_u + \|f'_{n_k} - f'_{n_l}\|_u + \|f'_{n_l} - f_{n_l}\|_u < 3\epsilon$$

Thus $\{f_{n_k}\}$ is uniformly Cauchy and hence convergent. Thus we conclude the claim by Bolzano-Weierstrass. □

Let $\epsilon > 0$. Following this claim, we pick finitely many $f_1, ..., f_n \in \overline{\mathcal{F}}$ such that $\overline{\mathcal{F}} \subset \bigcup_{i=1}^n B(f_i, \epsilon)$. Since $[0, 1]$ is compact, f_i is uniformly continuous for each i. Since there are only finitely many of them, we may take a small enough δ such that when $|x - y| < \delta$, $|f_i(x) - f_i(y)| < \epsilon$ for all i. Now let $f \in \overline{\mathcal{F}}$. There exists some i such that $f \in B(f_i, \epsilon)$. Then when $|x - y| < \delta$,

$$|f(x) - f(y)| \leq |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| < 3\epsilon$$

and since f is arbitrary we show that $\overline{\mathcal{F}}$ is equicontinuous are desired. In particular, $\mathcal{F}$ is equicontinuous. □

Problem 2.12. *Assume*

$$[0, 1] = \bigcup_{n=1}^{\infty} I_n$$

where $I_n = [a_n, b_n] \neq \emptyset$ and $I_m \cap I_n = \emptyset$ for $n \neq m$.

(a) Let $E = \{a_n : n \geq 1\} \cup \{b_n : n \geq 1\}$ be the set of endpoints. Prove E is closed.

(b) Prove no such family I_n exists.

Proof. 1. Notice that

$$[0, 1] = \bigcup_{n=1}^{\infty} I_n = \bigcup_{n=1}^{\infty} (I_n^\circ \cup \partial I_n) = \left(\bigcup_{n=1}^{\infty} I_n^\circ\right) \cup \left(\bigcup_{n=1}^{\infty} \partial I_n\right)$$

And that

$$\begin{aligned}\left(\bigcup_{n=1}^{\infty} I_n^\circ\right) \cap \left(\bigcup_{n=1}^{\infty} \partial I_n\right) &= \bigcup_{n=1}^{\infty} (\partial I_n \cap \bigcup_{n=1}^{\infty} I_n^\circ) \\ &= \bigcup_{n=1}^{\infty} \bigcup_{m=1}^{\infty} (\partial I_n \cap I_m^\circ) \\ &\subset \bigcup_{n=1}^{\infty} \bigcup_{m=1}^{\infty} (I_n \cap I_m) = \emptyset\end{aligned}$$

We get

$$E = \bigcup_{n=1}^{\infty} \partial I_n = \left(\bigcup_{n=1}^{\infty} I_n^\circ\right)^c$$

is closed since $\bigcup_{n=1}^{\infty} I_n^\circ$ is open.

2. Suppose on the contrary that such family exists. Since E is a closed subspace of $[0,1]$ compact, E is complete. For convenience, we denote $E_n := \partial I_n$. Then $E = \bigcup_1^\infty E_n$. We claim that each E_n is nowhere dense in E. Since E_n is closed, it suffices to show that E_n has empty interior. Let $x \in E_n$ and $r > 0$. Since $x \in \partial I_n$, $B(r,x)$ contains some y not in I_n, and hence $y \in I_m$ for some $m \neq n$. Thus $B(r,x)$ must contain $a_m \in E$ where $a_m \notin E_n$. Hence we show that x is a boundary point and E_n has empty interior. It follows that each E_n is nowhere dense. Since E is complete, it cannot be written as a countable union of E_n by Baire category theorem, and this gives a contradiction.

□

§3 Fall 2015

Problem 3.1. *Let $\{a_n\}_{n=1}^\infty$ be a sequence of positive integers such that*

$$a_{n+m} \leq a_n + a_m, \quad m, n \geq 1$$

Prove that $\lim_{n\to\infty} a_n/n$ exists by showing

$$\lim_{n\to\infty} \frac{a_n}{n} = \inf_{n\geq 1} \frac{a_n}{n}$$

Proof. Notice first that

$$\liminf_{n\to\infty} \frac{a_n}{n} := \sup_{k\geq 1}(\inf_{n\geq k} \frac{a_n}{n}) \geq \inf_{n\geq 1} \frac{a_n}{n}$$

Moreover, $\limsup_{n\to\infty} a_n/n := \inf_{k\geq 1}(\sup_{n\geq k} a_n/n)$ and for our convenience we call $C := \inf_{n\geq 1} a_n/n$. Let $\epsilon > 0$, there is some $k \in \mathbb{N}$ such that $a_k/k < C + \epsilon$. For the given ϵ, choose N large enough such that when $n \geq N$, $\max\{a_1, ..., a_{k-1}\}/n < \epsilon$. We claim that $\sup_{n\geq N} a_n/n \leq C + \epsilon$. Suppose $n \geq N$, $n = qk + r$ for some $r < k$ and hence by assumption

$$\frac{a_n}{n} \leq \frac{qa_k + a_r}{n} \leq \frac{a_k}{k} + \frac{a_r}{n} < C + \epsilon$$

and thus the claim is true. It follows that

$$\limsup_{n\to\infty} \frac{a_n}{n} := \inf_{k\geq 1}(\sup_{n\geq k} \frac{a_n}{n}) \leq \sup_{n\geq N} \frac{a_n}{n} \leq C + \epsilon = \inf_{n\geq 1} \frac{a_n}{n} + \epsilon$$

and thus $\limsup_{n\to\infty} a_n/n \leq \inf_{n\geq 1} a_n/n$ since ϵ is arbitrary. Then $\limsup_{n\to\infty} a_n/n \leq \inf_{n\geq 1} a_n/n \leq \liminf_{n\to\infty} a_n/n$ and we must have $\lim_{n\to\infty} a_n/n = \inf_{n\geq 1} a_n/n$, as desired. □

Problem 3.2. *Let $a, b \in \mathbb{R}$ obey $a < b$. Show that if $g, h : [a,b] \to \mathbb{R}$ are continuous with $h \geq 0$, then there is $c \in [a,b]$ such that*

$$\int_a^b g(x)h(x)dx = g(c)\int_a^b h(x)dx$$

Proof. Define $f(x) = g(x)\int_a^b h(t)dt$. It is clear that f is continuous as a continuous function g multiplied by a constant. Notice that g is continuous on $[a,b]$ compact, so by extreme value theorem, there exist x_0 and x_1 that are minimum and maximum of g, respectively. Since $h \geq 0$,

$$g(x_0)h(x) \leq g(x)h(x) \leq g(x_1)h(x)$$

for all x and integrating each side gives

$$\int_a^b g(x_0)h(x)dx \leq \int_a^b g(x)h(x)dx \leq \int_a^b g(x_1)h(x)dx \iff f(x_0) \leq \int_a^b g(x)h(x)dx \leq f(x_1)$$

Since f is continuous, by intermediate value theorem, there is some c such that $g(c)\int_a^b h(x)dx = f(c) = \int_a^b g(x)h(x)dx$. This finishes the proof. □

Problem 3.3. *Let $\{f_n\}$ be a sequence of continuous functions $f_n : [-1,1] \to [0,1]$ such that for each $x \in [-1,1]$,*

1. *The sequence of numbers $\{f(x)\}_{n=1}^{\infty}$ is non-increasing, and*
2. $\lim_{n\to\infty} f_n(x) = 0$.

Define

$$g_n(x) := \sum_{m=1}^{n} (-1)^m f_m(x)$$

Prove that $g_n(x)$ converges to some $g(x) \in \mathbb{R}$ for each $x \in [-1,1]$, and that the function $g : [-1,1] \to \mathbb{R}$ thus defined is continuous on $[-1,1]$.

Proof. On $[-1,1]$ compact, Since $\{f(x)\}_{n=1}^{\infty}$ is non-increasing for each x, f_n continous for each n, and $\{f_n\}$ pointwise converges to 0 which is continuous, $\{f_n\}$ uniformly converges to 0 on $[-1,1]$ (Rudin Theorem 7.13). Let $\epsilon > 0$, choose N large enough such that $\|f_n\|_u < \frac{\epsilon}{2}$ when $n > N$, then for $n, m > N$, where we assume $n < m$,

$$\begin{aligned}
|g_m(x) - g_n(x)| &= |\sum_{i=n}^{m} (-1)^i f_i(x)| = |(-1)^n f_n(x) + (-1)^{n+1} f_{n+1}(x) + ... + (-1)^m f_m(x)| \\
&\leq |f_n(x) + (f_{n+2}(x) - f_{n+1}(x)) + ... + (-1)^{m-1}(f_{m-1}(x) - f_m(x)))| \\
&\leq |f_n(x)| + |f_m(x)| < \epsilon \quad \text{(since all the middle terms cancel)}
\end{aligned}$$

for all x. Hence $\{g_n\}_n$ is uniformly Cauchy. Each g_n is easily observed to be continuous, and $C[-1,1]$ is complete under the uniform norm, so g_n uniformly converges to some g continuous on $[-1,1]$, as desired. □

Problem 3.4. *Let $f_n : [0,\infty) \to \mathbb{R}$ be functions defined recursively by $f_1(x) = 0$ and*

$$f_{n+1}(x) := e^{-2x} + \int_0^x f_n(t)e^{-2t}dt, \quad n \geq 1$$

Show that $f(x) := \lim_n f_n(x)$ exists for all $x \geq 0$ and identify f explicitly.

Proof Sketch. Define $V = \{f \in C[0,\infty) : \|f\|_u \leq 2\}$ and $Tf(x) = e^{-2x} + \int_0^x f(t)e^{-2t}dt$ an operator on V. First check that V is complete under the uniform norm, and then check that T is a contraction. Apply Banach fixed point theorem we get that there is some fixed point f that is the limit of this sequence. Eventually determine f by solving the integral equation

$$f(x) := e^{-2x} + \int_0^x f(t)e^{-2t}dt$$

□

Problem 3.5. *Let $F(x, y, z)$ be a continuously differentiable function with non-vanishing partial derivatives at $(0, 0, 0)$. Define functions*

$$x = x(y, z), \ y = y(x, z), \ \text{and } z = z(x, y)$$

as the solutions of the equation $F(x, y, z) = F(0, 0, 0)$ in the neighborhood of point $(0, 0)$ in the corresponding variables. Prove that

$$\frac{\partial x}{\partial y}\frac{\partial y}{\partial z}\frac{\partial z}{\partial x} = -1$$

where the three partial derivatives are taken at the point $(0, 0)$ in the corresponding pair of variables.

Proof. □

Problem 3.6. *Let $X = \mathbb{R}\backslash\{0\}$. Find a metric ρ on X with the following properties:*

1. *(X, ρ) is a complete metric space, and*
2. *If $\{x_n\}_1^\infty \subset X$ and $x \in X$, then*

$$\lim_{n\to\infty} |x_n - x| = 0 \iff x_n \to x \text{ in}(X, \rho)$$

Prove both properties, as well as all your other assertions, in full detail.

Proof Sketch. Consider the metric

$$\rho(x, y) = |x - y| + \frac{1}{|xy|}$$

and check that it satisfies the desired properties. □

Problem 3.7. *Let A be a $n \times n$ real matrix such that $A^T = -A$. Prove that* $\det(A) \geq 0$.

Proof. Suppose λ is an eigenvalue of A, and we pick an eigenvector v corresponding to λ. Then

$$\lambda\langle v, v\rangle = \langle \lambda v, v\rangle = \langle Av, v\rangle = \langle v, A^T v\rangle = \langle v, -\lambda v\rangle = -\overline{\lambda}\langle v, v\rangle$$

meaning that all eigenvalues are imaginary. Moreover, since A is real, $\det(\lambda I - A)$ is a real polynomial. Taking conjugation tells us that if λ is an eigenvalue so is $\overline{\lambda}$. Two facts combined we get that $\det(A) \geq 0$ as the product of all eigenvalues. □

Problem 3.8. *Let A, B be two 4×5 matrics of rank 3, and let $C = A^T B$ (this is a 5×5 matrix. Find all possible values r for the rank of C. To be precise, if the rank r is impossible, find an explicit example of such matrices. Then prove that all other values are impossible.*

Proof. The only possible ranks are 2 and 3. We first show that all other ranks are impossible. $\dim \operatorname{Im}(A^T B) \leq \max(\dim \operatorname{Im} A^T, \dim \operatorname{Im} B) = 3$, all ranks beyond 3 are impossible. On the other hand, define $W = \operatorname{Im} B$, and thus W is a 3-dimensional subspace of $\mathbb{R}^4$. Consider the linear map $A^T\ |_W \colon W \to \mathbb{R}^5$. $\dim \ker A^T\ |_W \leq \dim \ker A^T = 1$, so by rank-nullity $\dim \operatorname{Im} A^T\ |_W \geq 2$. Since $\operatorname{Im} A^T B = \operatorname{Im} A^T\ |_W$, we have $\dim \operatorname{Im} A^T B \geq 2$, as desired. Thus

$$2 \leq \dim \operatorname{Im} A^T B \leq 3$$

We then use specific examples to illustrate rank 2 and 3 are both possible. For rank 2, consider

$$A^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Then

$$A^T B = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

has rank 2.

For rank 3, consider

$$A^T = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Then

$$A^T B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

has rank 3. □

Problem 3.9. *Find M^{-2}, where*

$$M = \begin{pmatrix} 2 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 4 & 8 & 6 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}$$

Proof. To brute force it, we can first find M^{-1} by juxtaposing M and the identity matrix and row reduce this augmented matrix, which gives

$$M^{-1} = \begin{pmatrix} -4 & -1 & 0 & 0 \\ 5 & 2 & 0 & 0 \\ -6 & -2 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{pmatrix}$$

And hence

$$M^{-2} = \begin{pmatrix} 11 & 2 & 0 & 0 \\ -10 & -1 & 0 & 0 \\ 14 & 2 & 1 & 2 \\ -6 & -2 & -2 & -3 \end{pmatrix}$$

as desired. □

Problem 3.10. *Let $F, G : \mathbb{R}^n \to \mathbb{R}^n$ be two linear operators. Recall that*

$$\exp(F) = \sum_{k=0}^{\infty} \frac{1}{k!} F^k$$

1. *Prove that when F and G are commuting, i.e. $FG = GF$, we have*

$$\exp(F + G) = \exp(F)\exp(G)$$

2. *Give an example of non-commuting linear operators when this equality fails.*

Proof. 1. The gist is that since F and G commute, then the operation is completely analogous to the case of exponential of real numbers. So all properties of $e^x : \mathbb{R} \to \mathbb{R}$ should carry to the matrix situation.

2. Let

$$F = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad G = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Then

$$\exp(F) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \exp(G) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

since $0 = F^2 = G^2$. Then

$$\exp(F)\exp(G) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

Meanwhile, notice that $(F + G)^n$ have positive entries for all n, and hence

$$\begin{aligned} \exp(F + G) &= I + (F + G) + \frac{(F + G)^2}{2} + \dots \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} + \dots \end{aligned}$$

cannot be equal to $\exp(F)\exp(G)$ since the last entry of $\exp(F + G)$ must be > 1.

□

Problem 3.11. *Let $T : V \to V$ be a linear operator such that $T^6 = 0$ and $T^5 \neq 0$. Suppose $V \cong \mathbb{R}^6$. Prove that there is no linear operator $S : V \to V$ such that $S^2 = T$. Does the answer change if $V \cong \mathbb{R}^{12}$?*

Proof. Suppose there is some $S : V \to V$ such that $S^2 = T$. Then $S^{12} = T^6 = 0$ meaning that S is nilpotent. But then $S^6 = 0$, meaning that $S^{10} = T^5 = 0$, a contradiction. Hence there is no operator S such that $S^2 = T$. If $V \cong \mathbb{R}^{12}$. This will be possible. Consider the 12×12 matrix with 0 as the only eigenvalue and has only 1 Jordan block. Then the minimal polynomial is x^12, showing that $T^6 = S^{12} = 0$ and $T^5 = S^{10} \neq 0$, as desired. □

Problem 3.12. *Prove that the following $n \times n$ matrix M is positive definite:*

$$M = \begin{pmatrix} 2 & 1 & 1 & \dots & 1 \\ 1 & 3 & 1 & \dots & 1 \\ 1 & 1 & 4 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \dots & n+1. \end{pmatrix}$$

Proof Sketch. Since M is symmetric, by Sylvester's criterion, it suffices to show that every $k \times k$ upper left block matrix has determinant > 0, which is a formal check. □

§4 Spring 2015

Problem 4.1. *Let $f : [0, \infty) \to [0, \infty)$ be continuous with $f(0) = 0$. Show that if*

$$f(t) \leq 1 + \frac{1}{10} f(t)^2$$

then f is uniformly bounded throughout $[0, \infty)$.

Proof. Rewrite the above formula as

$$f(t)^2 - 10f(t) + 10 \geq 0$$

Notice that

$$t^2 - 10t + 10 \geq 0$$

when $t \in (-\infty, \frac{10-\sqrt{60}}{2}] \cup [\frac{10+\sqrt{60}}{2}], \infty)$, where the range of f also lies in. Since f takes 0 as a possible value and the range of f is connected, the range of f lies in $(-\infty, \frac{10-\sqrt{60}}{2}]$. Therefore f is uniformly bounded. □

Problem 4.2. *Let $f : [0, 1] \to \mathbb{R}$. We say that f is Holder continuous of order $\alpha \in (0, 1)$ and write $f \in C^\alpha[0, 1]$ if*

$$\|f\|_{c^\alpha} := \sup\{|f(x)| : x \in [0, 1]\} + \sup\{\frac{|f(x) - f(y)|}{|x - y|^\alpha} : x \neq y \in [0, 1]\} < \infty$$

This defines a norm on $C^\alpha[0, 1]$. Prove that any bounded sequence in $C^{1/2}[0, 1]$ admits a convergent subsequence in $C^{1/3}[0, 1]$.

Proof. Notice that

$$\frac{|f(x) - f(y)|}{|x - y|^{1/3}} = |x - y|^{1/6} \frac{|f(x) - f(y)|}{|x - y|^{1/2}} \leq \frac{|f(x) - f(y)|}{|x - y|^{1/2}}$$

and hence $\|f\|_{C^{1/3}} \leq \|f\|_{C^{1/2}}$. Thus a bounded sequence $\{f_n\}_n \subset C^{1/2}[0, 1]$ is also a bounded sequence $C^{1/3}[0, 1]$. By bounded convergence theorem, $\{f_n\}_n$ admits a convergent subsequence in $C^{1/3}[0, 1]$, as desired.

□

Problem 4.3. *Let $f : \mathbb{R} \to \mathbb{R}$ be a Lipschitz function. Suppose that for every $x \in \mathbb{R}$,*

$$\lim_{n\to\infty} n \left(f\left(x + \frac{1}{n}\right) - f(x) \right) = 0$$

Prove that f is differentiable.

Proof. Suppose L is the Lipschitz constant of f and let $\varepsilon > 0$. Choose n sufficiently large so that $n > \dfrac{4}{\varepsilon L}$ and

$$n \left| f\left(x + \frac{1}{n}\right) - f(x) \right| < \frac{\varepsilon}{2}.$$

If $\frac{1}{n+1} \leq h \leq \frac{1}{n}$, then

$$\begin{aligned}
\left|\frac{f(x+h)-f(x)}{h}\right| &\leq \left|\frac{f(x+h)-f\left(x+\frac{1}{n}\right)}{h}\right| + \left|\frac{f\left(x+\frac{1}{n}\right)-f(x)}{h}\right| \\
&\leq \frac{L\left|h-\frac{1}{n}\right|}{h} + \frac{\left|f\left(x+\frac{1}{n}\right)-f(x)\right|}{h} \\
&\leq L(n+1)\frac{1}{n(n+1)} + (n+1)\left|f\left(x+\frac{1}{n}\right)-f(x)\right| \\
&\leq \frac{L}{n} + \frac{\varepsilon}{2} + \frac{L}{n} \\
&< \varepsilon.
\end{aligned}$$

This shows that $\lim_{h\to 0}\left|\frac{f(x+h)-f(x)}{h}\right| = 0$, and hence (by the squeeze theorem) $f'(x) = 0$ for all $x \in \mathbb{R}$.

□

Problem 4.4. *Let $f : [0,1] \to \mathbb{R}$ be a function satisfying the intermediate value property, namely, whenever $0 \leq a < b \leq 1$ and y lies between $f(a)$ and $f(b)$, there exists $x \in (a,b)$ such that $f(x) = y$. Assume that for any $y \in \mathbb{R}$, the preimage $f^{-1}(y)$ is closed. Prove that f is continuous.*

Proof. Assume for sake of contradiction that f is not continuous at $a \in [0,1]$. Thus there exists an $\varepsilon > 0$ such that for all natural numbers $n > 1$, there exists an $x_n \in [0,1]$ such that $|f(x_n) - f(a)| \geq \varepsilon$ and $|x_n - a| < \frac{1}{n}$. WLOG we can pass to a subsequence (x_{n_k}) so that $f(x_{n_k}) > f(a)$ for all k. If $f(x_n) > f(a)$ for only finitely many n then choose the subsequence such that $f(x_{n_k}) < f(a)$ instead, and the proof will work similarly.

Fix $y \in (f(a), f(a)+\varepsilon)$. Since $y \in (f(a), f(x_{n_k}))$ for all k, using the intermediate value property we obtain a sequence $(t_k) \subset [0,1]$ such that $f(t_k) = y$ and $|t_k - a| < \dfrac{1}{n_k}$ for all k. It follows that a is a limit point of $f^{-1}(y)$ which is closed, so that $a \in f^{-1}(y)$. But then $f(a) = y$, a contradiction. □

Problem 4.5. *Let $f : [1,\infty) \to [0,\infty)$ be bounded and monotonically decreasing with $\lim_{x\to\infty} f(x) = 0$. Show that*

$$\int_1^{N+1} f(x)dx - \sum_{n=1}^{N} f(n)$$

converges to a finite limit as $N \to \infty$.

Proof. Since f is monotonically decreasing, for each natural number n we have

$$f(n+1) \leq \int_n^{n+1} f(x)dx \leq f(n).$$

Hence

$$\sum_{n=1}^{N} f(n+1) - f(n) \leq \int_1^{N+1} f(x)dx - \sum_{n=1}^{N} f(n).$$

The left hand side sum telescopes to $f(N) - f(1)$ which approaches $-f(1)$ as $N \to \infty$. Note that $S_N = \int_1^{N+1} f(x)dx - \sum_{n=1}^{N} f(n)$ is decreasing since

$$S_N - S_{N-1} = \int_N^{N+1} f(x)dx - f(N) \leq 0.$$

S_N is bounded below by $-f(1)$ so $\lim_{N\to\infty} S_N = \inf S_N$ by the monotone convergence theorem. □

Problem 4.6. *Prove that the integral equation*

$$f(t) = e^{t^2} + \frac{1}{2}\int_0^1 \cos(s)f(s)ds$$

admits a unique continuous solution $f : [0,1] \to \mathbb{R}$.

Proof. For convenience, define $V := C[0,1]$. Define an operator $T : V \to V$ such that

$$T(f)(t) = e^{t^2} + \frac{1}{2}\int_0^1 \cos(s)f(s)ds$$

Then for $f, g \in V$,

$$\begin{aligned} |T(f)(t) - T(g)(t)| &= |\frac{1}{2}\int_0^1 \cos(s)(f(s) - g(s))ds| \\ &\leq \frac{1}{2}\int_0^1 |\cos(s)||f(s) - g(s)|ds \\ &\leq \frac{1}{2}\|f - g\|_u \end{aligned}$$

and hence

$$\|T(f) - T(g)\|_u \leq \frac{1}{2}\|f - g\|_u$$

Since V is a Banach space under the uniform norm, we conclude by contraction mapping theorem that T admits a unique fixed point in V, which solves the equation. □

Remark 4.7. *With a little bit more effort one can show that*

$$f(t) = e^{t^2} + \frac{\int_0^1 \cos(s)e^{s^2}ds}{2 - \sin(1)}$$

is the unique solution.

Problem 4.8. *Let*

$$f(x,y,z) = 9x^2 + 6y^2 + 6z^2 + 12xy - 10xz - 2yz$$

Does there exist a point (x,y,z) *such that* $f(x,y,z) < 0$*?*

Proof. The answer is no. Define

$$M = \begin{pmatrix} 9 & 6 & 5 \\ 6 & 6 & -1 \\ -5 & -1 & 6 \end{pmatrix}$$

Then it is easy to verify $f(x) = x^T M x$ for $x \in \mathbb{R}^3$. Notice that

$$\det \begin{pmatrix} 9 & 6 \\ 6 & 6 \end{pmatrix} = 18 > 0, \quad \det M = \det \begin{pmatrix} 9 & 6 & 5 \\ 6 & 6 & -1 \\ -5 & -1 & 6 \end{pmatrix} = 9 > 0$$

and M is symmetric. By Sylvester's criterion for positive-definiteness, we obtain that M is positive definite and hence $f(x) \geq 0$. Hence there can be no $x \in \mathbb{R}^3$ such that $f(x) < 0$. □

Problem 4.9. *Prove or disprove the following statements:*

1. *Matrices with determinant 1 are dense in the set of* 3×3 *matrices.*
2. *Matrices with distinct eigenvalues are dense in the set of all* 3×3 *complex matrices.*

Proof. 1. The conclusion is false. We claim that determinant is a continuous function $M_3(\mathbb{R}) \to \mathbb{R}$. To show this claim, we first notice that $M_3(\mathbb{R})$ is a metric space since it can be identified with $\mathbb{R}^9$. Therefore, it suffices to show that for $A \in M_3(\mathbb{R})$ with $A_n \to A$, $\det(A_n) \to \det(A)$. The topology here is pointwise convergence, so $A_n \to A$ means that every entry of A_n converges to their corresponding entry in A as $n \to \infty$. But $\det(A)$ is in particular a polynomial with variables the entries of A, sending $n \to \infty$ and noting that limit is multiplicative and additive we get that $\det A_n \to \det A$. It follows immediately from the claim that matrices with determinant 1 cannot be dense.

2. The conclusion is true. Let A be a matrix and $A = SJS^{-1}$ be the Jordan decomposition of A. The idea is to perturb J such that it becomes an upper triangular matrix with distinct diagonal entries and using the fact that SMS^{-1} is a continuous function $M_3(\mathbb{C}) \to M_3(\mathbb{C})$. □

Problem 4.10. *Let* $V = \mathbb{R}^n$ *and* U_1, U_2, W_1, W_2 *be subspaces of* V *of dimension* d*, such that* $\dim(U_1 \cap W_1) = \dim(U_2 \cap W_2) = l$, $l \leq d \leq n$. *Prove that there exists a linear operator* $T : V \to V$ *such that* $T(U_1) = U_2$ *and* $T(W_1) = W_2$.

Proof. Suppose $\{x_1, ..., x_l\}$ is a basis of $U_1 \cap W_1$, and $\{y_1, ..., y_l\}$ is a basis of $U_2 \cap W_2$. We extend to $\{x_1, ..., x_l, u_1, ..., u_{d-l}\}$ a basis of U_1, $\{x_1, ..., x_l, u'_1, ..., u'_{d-l}\}$ a basis of U_2. Then define T such that T fixes x_i and $T(u_i) = u'_i$. Extending by linearity we know that such T is uniquely defined, and the definition tells that T is bijective. Then we pick a basis $\{x_1, ..., x_l, w_1, ..., w_{d-l}\}$ a basis of W_1, and we claim that $\{x_1, ..., x_l, T(w_1), ..., T(w_{d-l})\}$ is a basis of W_2. Since the size of this set equals the dimension of W_2, it suffices to show that it is linearly independent. Suppose

$$a_1 x_1 + ... a_l x_l + b_1 T(w_1) + ... + b_{d-l} T(w_{d-l}) = 0$$

by our definition of T we have

$$T(a_1 x_1 + ... + a_l x_l + b_1 w_1 + ... + b_{d-l} w_{d-l}) = 0$$

and by bijectivity

$$a_1 x_1 + ... + a_l x_l + b_1 w_1 + ... + b_{d-l} w_{d-l} = 0$$

and thus $a_1 = ... = a_l = b_1 = ... = b_{d-l} = 0$. Thus we show linearly independence as desired. □

Problem 4.11. *Let*

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad M^{-1} = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$$

where $A, ..., S$ are $k \times k$ matrices. Show that

$$\det M \cdot \det S = \det A$$

Proof. We first suppose A is invertible. Since $MM^{-1} = I$, we have the following:

$$AP + BR = I, \; CP + DR = 0$$
$$AQ + BS = 0, \; CQ + DS = I$$

Then $Q = -A^{-1}BS$ and $(D - CA^{-1}B)S = I$. It follows that

$$1 = \det(I) = \det(S)\det(D - CA^{-1}B) = \det(M)/\det(A)$$

and thus the conclusion holds. Now we claim that

$$\mathcal{A} := \{M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in M_k(\mathbb{C}) \mid M \text{ is invertible}\}$$

and

$$\mathcal{B} := \{M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in M_k(\mathbb{C}) \mid A \text{ is invertible}\}$$

are both open dense subsets. Suppose the claim holds, Baire category theorem implies that $\mathcal{A} \cap \mathcal{B}$ is dense in $M_k(\mathbb{C})$. We then pick $\{M_j\}_j \subset \mathcal{A} \cap \mathcal{B}$ such that $M_j \to M$, and denote

$$M_j = \begin{pmatrix} A_j & B_j \\ C_j & D_j \end{pmatrix}, \quad M_j^{-1} = \begin{pmatrix} P_j & Q_j \\ R_j & S_j \end{pmatrix}$$

By continuity of the determinant function,

$$\det(A) = \lim_{j\to\infty} \det(A_j) = \lim_{j\to\infty} \det(M_j) \cdot \det(S_j) = \det(M)\det(S)$$

as desired. We now prove the claim. That $\mathcal{A}$ is dense is a standard result, and to see that $\mathcal{B}$ is dense, suppose

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in M_k(\mathbb{C})$$

and define

$$M_j = \begin{pmatrix} A_j & B \\ C & D \end{pmatrix} \in \mathcal{B}$$

such that $A_j \to A$, which is possible by the density of $\mathcal{A}$. Then clearly $M_j \to M$, showing denseness of $\mathcal{B}$. To see that $\mathcal{A}$ and $\mathcal{B}$ are open, consider the continuous functions $f : M \mapsto \det(M)$ and $g : M \mapsto \det(A)$. $\mathcal{A} = f^{-1}(\mathbb{C}\backslash\{0\})$ and $\mathcal{B} = g^{-1}(\mathbb{C}\backslash\{0\})$ are hence open, as desired. This finishes the proof. □

Problem 4.12. *Let*

$$M = \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix}$$

(a) *Compute* $\exp(M)$.

(b) Does there exist a real 2×2 matrix A such that $M = \exp(A)$?

Proof. (a) We write $M = SDS^{-1}$, where

$$S = \begin{pmatrix} 1 & 5 \\ -1 & 1 \end{pmatrix}, \ D = \begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix}, \ S^{-1} = \frac{1}{6}\begin{pmatrix} 1 & -5 \\ 1 & 1 \end{pmatrix}$$

and hence

$$\exp(M) = S[\exp(D)]S^{-1} = \frac{1}{6}\begin{pmatrix} 1 & 5 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} e^{-2} & 0 \\ 0 & e^4 \end{pmatrix}\begin{pmatrix} 1 & -5 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{e^{-2}+5e^4}{6} & \frac{-5e^{-2}+5e^4}{6} \\ \frac{-e^{-2}+e^4}{6} & \frac{5e^{-2}+e^4}{6} \end{pmatrix}$$

(b) Suppose $M = \exp(A)$ for some 2×2 matrix A. We denote the Jordan decomposition as $A = SJS^{-1}$. If A only has one eigenvalue λ, then

$$M = \exp(A) = S[\exp(J)]S^{-1} = S\begin{pmatrix} e^{\lambda} & * \\ 0 & e^{\lambda} \end{pmatrix} S^{-1}$$

showing that M would have only one eigenvalue, a contradiction. Thus J must have the form

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

and

$$M = \exp(A) = S[\exp(J)]S^{-1} = S\begin{pmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{pmatrix} S^{-1}$$

showing that M has eigenvalues e^{λ_1} and e^{λ_2}, both positive. This is a contradiction, and hence M cannot be the exponential of any 2×2 matrix.

□

§5 Fall 2016

Problem 5.1. *Suppose A,B are invertible $n \times n$ matrices with complex entries such that $ABA^{-1} = B^3$. Show that all eigenvalues of B are roots of unity.*

Proof. Since B and B^3 are similar, they have the same eigenvalues. Hence if λ is an eigenvalue of B, λ^3, being an eigenvalues of B^3, is also an eigenvalues of B. Then λ^9 is an eigenvalues of B^3 and hence again an eigenvalue of B. Iteratively we get that λ^{3k} are all eigenvalues of B and by Pigeonhole principle $\lambda^{3k} = \lambda$ for some k. Since B is invertible, $\lambda \neq 0$. Thus $\lambda^{3k-1} = 1$, showing that λ is a root of unity. □

Problem 5.2. *Suppose*

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Find the smallest possible constant $C > 0$ so that

$$||e^A x|| \leq C||x||$$

for all $x \in \mathbb{R}^2$. $||\cdot||$ denotes the Euclidean norm.

Proof. We calculate

$$e^A = \exp\left(\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} e^2 & 0 \\ 0 & e^2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} e^2 & e^2 \\ 0 & e^2 \end{pmatrix}.$$

Now to find C we compute the operator norm $||e^A||_{op} = \sup_{x\in\mathbb{R}^2} \dfrac{||e^A x||}{||x||} = \sup_{||x||=1} ||e^A x||$. Let $x = (x_1, x_2)^T$ have unit length. Then

$$||e^A x|| = e^2 \cdot ||(x_1, x_2) + (x_2, x_2)|| \leq e^2(1 + \sqrt{2}).$$

□

Problem 5.3. *Let B be a 7×7 matrix and A_n a sequence of matrices of the same size. Suppose that*

$$A_n \to B \quad as \quad n \to \infty,$$

in the sense that the individual matrix entries converge. Show that

$$\operatorname{rank} B \leq \liminf_{n\to\infty} \operatorname{rank} A_n.$$

Proof. This follows from lower semicontinuity of rank. That is, let $A : X \to M_{n\times n}$ be continuous where X is a topological space. Then $x \mapsto A(x)$ is lower semincontinuous.
To prove this, let $a \in \mathbb{R}$ and $x_0 \in (\operatorname{rank} A)^{-1}(a, \infty)$. Then $k = \operatorname{rank} A(x_0)) > a$. It follows there exists some $k \times k$ minor of $A(x_0)$, say $B(x_0)$, with nonzero determinant.$x \mapsto \det B(x)$ is a continuous function, so there exists an open neighborhood U of x_0 on which $\det B(x) \neq 0$. Therefore $\operatorname{rank}(A(U)) \subset [k, \infty) \subset (a, \infty)$. Hence $(\operatorname{rank} A)^{-1}(a, \infty)$ is open.

Specializing to the problem, if $r = \operatorname{rank} B$ and $\epsilon > 0$ then $\operatorname{rank}^{-1}(r-\epsilon, \infty)$ is open. Since rank is integer valued we see there exists a neighborhood U of B such that $\operatorname{rank} A \geq \operatorname{rank} B$ for all $A \in U$. As $A_n \to B$, it follows that $\liminf_{n\to\infty} \operatorname{rank} A_n \geq \operatorname{rank} B$, as desired. □

Problem 5.4. *Let $\alpha_1, \ldots, \alpha_n$ be complex numbers, and V be the vector subspace of $\mathbb{C}$ spanned by them over the field of rational numbers, $\mathbb{Q}$. Let β be a complex number such that $\beta V \subset V$. Show that β is a root of a degree n polyomial with rational coefficients.*

Proof. For each $i = 1, \ldots, n$ we can write $\beta\alpha_i = \sum_{j=1}^n r_{ij}\alpha_j$ for $r_{ij} \in \mathbb{Q}$. Write $\alpha = (\alpha_1, \ldots, \alpha_n)^T$ and $R = (r_{ij})$. It follows that $R\alpha = \beta\alpha$, so β is an eigenvalue of R. Therefore β is a root of the characteristic polynomial of R, whose coefficients are polynomials in the r_{ij}'s and hence rational. □

Problem 5.5. *let A be an invertible $n \times n$ matrix whose entries belong to $\mathbb{Q}$. Suppose that $Av = 3v$ for a vector $v \in \mathbb{R}^n$.*

1. *Give an example in which no non-zero scalar multiple λv for $\lambda \in \mathbb{R}$ is in $\mathbb{Q}^n$.*
2. *Show that there is always a non-zero $w \in \mathbb{Q}^n$ such that $Aw = 3w$.*

Proof. 1. Consider $v = (1, \sqrt{2})^T$. 1 and $\sqrt{2}$ are $\mathbb{Q}$-linearly independent, and hence no non-zero λ can make $\lambda v \in \mathbb{Q}^n$.

2. The assumption that $Av = 3v$ for some $v \in \mathbb{R}^n$ means that $(A - 3I)x = 0$ has a non-trivial solution in $\mathbb{R}^n$. This means that doing Gaussian elimination we can row reduce $A - 3I$ to specific form (the rank is strictly less than n). Since $A - 3I$ has rational entries, the process of row-reduction is conducted entirely in $\mathbb{Q}$. Hence regarding us as solving the equation in $\mathbb{Q}^n$ we see that $A - 3I$ has a can be row-reduced to a matrix with rank less than n and has a solution in $\mathbb{Q}^n$. Any particular solution, denoted w, would satisfy $Aw = 3w$.

□

Problem 5.6. *Determine the volume of the region in $\mathbb{R}^3$ determined by the following inequalities:*

$$x^2 + y^2 + z^2 \leq 4, \qquad x^2 - 2x + y^2 + z^2 \geq 0, \quad \textit{and} \quad z \geq x.$$

Proof. The desired volume is the volume bound by the sphere of radius 2 centered at the origin and the plane $z = x$, minus the volume of the spherical cap bound by $z = x$ and the sphere of radius 1 centered at $(1, 0, 0)$. We caluculate the volume of the spherical cap as a volume of revolution:

$$\pi \int_0^{1-\sqrt{2}/2} 1 - (t-1)^2 dt = \pi \left(1 - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{12} - \frac{1}{3}\right) = \frac{\pi(8 - 5\sqrt{2})}{12}.$$

So the desired volume is $16\pi - \dfrac{\pi(8 - 5\sqrt{2})}{12}$. □

Problem 5.7. *A function $f : \mathbb{R} \to \mathbb{R}$ is* convex *iff for all $x, y \in \mathbb{R}$ and $t \in [0, 1]$,*

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y). \tag{1}$$

Suppose that f is differentiable. Prove that f is convex iff

$$f(y) \geq f(x) + (y - x)f'(x) \tag{2}$$

for all $x, y \in \mathbb{R}$.

Proof. Suppose f is convex. Then rearranging (1) we get

$$f(y) \geq f(x) + \frac{f(x + (y-x)t) - f(x)}{t}.$$

As $t \to 0$ we get $f(y) \geq f(x) + f'(x)(y - x)$ as desired.
Conversely suppose (2) holds. Let $x, y \in \mathbb{R}$ be distinct and let z be some point in between i.e. $z = \alpha x + (1 - \alpha)y$ for some $\alpha \in [0, 1]$. Then

$$f(x) \geq f(z) + f'(z)(x - z)$$

and similarly

$$f(y) \geq f(z) + f'(z)(y - z).$$

Hence,

$$\begin{aligned} \alpha f(x) + (1-\alpha)f(y) &\geq \alpha f(z) + \alpha f'(z)(x - z) + (1-\alpha)f(z)(y - z) \\ &= f(z) + f'(z)(\alpha(1-\alpha)(x - y) + (1-\alpha)\alpha(y - x)) \\ &= f(z). \end{aligned}$$

Therefore f is convex, as desired. □

§6 Spring 2016

Problem 6.1. *For $a < b$, let $f : [a, b] \times [a, b] \to \mathbb{R}$ be such that*

(a) for each $y \in [a, b]$, $x \mapsto f(x, y)$ is non-increasing and continuous on $[a, b]$.

(b) for each $x \in [a, b]$, $y \mapsto f(x, y)$ is non-decreasing and continuous on $[a, b]$.

Prove that $g(x) = f(x, x)$ is continuous on $[a, b]$.

Proof. Let $\epsilon > 0$ and choose δ such that $f(x, x + \delta) - f(x, x) < \epsilon/2$. By continuity there exists $\delta' < \delta$ such that $f(x, x+\delta) - f(x+\delta', x+\delta) < \epsilon/2$. Since $f(x+\delta', x+\delta) \geq f(x+\delta', x+\delta')$, we get

$$f(x, x + \delta) - f(x + \delta', x + \delta') \leq f(x, x + \delta) - f(x + \delta', x + \delta').$$

And hence, $|g(x) - g(x + \delta')| \leq |g(x) - f(x, x + \delta)| + |f(x, x + \delta) - g(\delta')| < \epsilon$. This shows that g is continuous from above. A symmetrical argument will show continuity from below. □

Problem 6.2. *For $a < b$ real numbers and $f : [a, b] \to \mathbb{R}$ a function*

1. *Define what it means for f to be Riemann integrable on $[a, b]$.*
2. *Let $\{x_n\} \subset [a, b]$ be a seqeunce such that $\lim_{n\to\infty} x_n$ exists and suppose that $f : [a, b] \to \mathbb{R}$ is defined by*

$$f(x) = \begin{cases} 1 & \text{if } x \notin \{x_n\} \\ 0 & \text{else .} \end{cases}$$

Using your definition, prove f is Riemann integrable.

Proof. We just do 2. Let $\xi \in [a, b]$ be the limit of the sequence. Let $\epsilon > 0$ and choose N large enough so that $x_n \in (\xi - \epsilon, \xi + \epsilon)$ for all $n > N$. For $n = 1, 2, \ldots, N$ cover x_n with intervals $(x_n - \epsilon, x_n + \epsilon)$. We can assume ϵ is sufficiently small so that all these intervals are contained in $[a, b]$, with the caveat that we use half-closed intervals $[a$, Construct a partition $P = \{a = y_0 < y_1 < \cdots < y_n = b\}$ of $[a, b]$ containing the endpoints of these intervals, together with $\xi \pm \epsilon$. Clearly the upper Riemann sum satisfies $U(f, P) = b - a$ since the points where $f(x) = 1$ are dense. Moreover, the lower Riemann sum satisfies

$$L(f, P) = \sum_{n=0}^{k-1} \inf_{x\in[y_i,y_{i+1}]} f(x) \cdot (y_{i+1} - y_i) \geq (b - a) - (N + 1) \cdot 2\epsilon.$$

As ϵ was arbitrary we conclude f is Riemann integrable and $\int_a^b f = b - a$. □

Problem 6.3. *Suppose $f : [0, 1] \to \mathbb{R}$ is a continuously differentiable function. Show that the limit*

$$\lim_{n\to\infty} n\left(\sum_{k=0}^{n} f\left(\frac{k}{n}\right) - n\int_0^1 f(x)dx\right)$$

exists and compute its value.

Remark 6.4. *This problem surely contains a typo. Suppose we have $f(x) = x$, then $f \in C^1([0,1])$. However it is easy to see that the limit blows up to ∞. The intended question is probably the following:*

Suppose $f : [0,1] \to \mathbb{R}$ is a continuously differentiable function. Show that the limit

$$\lim_{n\to\infty} \sum_{k=0}^{n} f\left(\frac{k}{n}\right) - n\int_0^1 f(x)dx$$

exists and compute its value.

Proof. Let $\epsilon > 0$. By uniform continuity of f and f', there exists a $\delta > 0$ such that when $|x-y| < \delta$, $|f(x) - f(y)| < \epsilon$ and $|f'(x) - f'(y)| < \epsilon$. Take N large enough such that when $n \geq N$, $\frac{1}{n} < \delta$. Fix some $n \geq N$. Then

$$\int_0^1 f(x)dx = \sum_{k=0}^{n-1} \int_{\frac{k}{n}}^{\frac{k+1}{n}} f(x)dx = \sum_{k=0}^{n-1} \int_0^{\frac{1}{n}} f(\frac{k}{n} + l)dl$$

By Mean Value Theorem, $f(\frac{k}{n} + l) = f\left(\frac{k}{n}\right) + lf'(c_k)$ for some $c_k \in [\frac{k}{n}, \frac{k}{n} + l]$. Moreover, $f'(c_k) = f'\left(\frac{k}{n}\right) + \Delta_k(l)$, where $|\Delta_k(l)| < \epsilon$. Substitution gives

$$f\left(\frac{k}{n} + l\right) = f\left(\frac{k}{n}\right) + l(f'\left(\frac{k}{n}\right) + \Delta_k(l))$$

and thus

$$\int_0^{\frac{1}{n}} f\left(\frac{k}{n} + l\right) dl = \frac{1}{n} f\left(\frac{k}{n}\right) + \frac{1}{2n^2} f'\left(\frac{k}{n}\right) + \int_0^{\frac{1}{n}} l\Delta_k(l)dl$$

On the other hand, we have

$$f\left(\frac{k}{n} - l\right) = f\left(\frac{k}{n}\right) - lf'(c_k') = f\left(\frac{k}{n}\right) - l(f'\left(\frac{k}{n}\right) + \Delta_k'(l))$$

where $|\Delta_k'(l)| < \epsilon$. Thus

$$\int_0^{\frac{1}{n}} f\left(\frac{k}{n} - l\right) dl = \frac{1}{n} f\left(\frac{k}{n}\right) - \frac{1}{2n^2} f'\left(\frac{k}{n}\right) - \int_0^{\frac{1}{n}} l\Delta_k'(l)dl$$

Then

$$\int_0^1 f(x)dx = \sum_{k=0}^{n-1} \int_{\frac{k}{n}}^{\frac{k+1}{n}} f(x)dx = \sum_{k=0}^{n-1} \left[\frac{1}{n} f\left(\frac{k}{n}\right) + \frac{1}{2n^2} f'\left(\frac{k}{n}\right) + \int_0^{\frac{1}{n}} l\Delta_k(l)dl\right] \quad (1)$$

and

$$\int_0^1 f(x)dx = \sum_{k=1}^{n} \int_{\frac{k}{n}}^{\frac{k+1}{n}} f(x)dx = \sum_{k=1}^{n} \left[\frac{1}{n} f\left(\frac{k}{n}\right) - \frac{1}{2n^2} f'\left(\frac{k}{n}\right) - \int_0^{\frac{1}{n}} l\Delta_k'(l)dl\right] \quad (2)$$

Adding (1) and (2) together and rearranging the terms we get

$$\begin{aligned}
&|2\int_0^1 f(x)dx - \sum_{k=1}^{n} \frac{2}{n} f\left(\frac{k}{n}\right) - f(0) - f(1)| \\
\leq &|\frac{1}{2n^2} f'(0) - \frac{1}{2n^2} f'(1)| + \sum_{k=0}^{n-1} |\int_0^{\frac{1}{n}} l\Delta_k(l)dl| + \sum_{k=1}^{n} |\int_0^{\frac{1}{n}} l\Delta_k'(l)dl| \\
\leq &|\frac{1}{2n^2} f'(0) - \frac{1}{2n^2} f'(1)| + \frac{\epsilon}{n}
\end{aligned}$$

where the last step uses the estimate that $|\Delta_k|$ and $|\Delta'_k|$ are both less than ϵ. It follows that

$$|n\int_0^1 f(x)dx - \sum_{k=1}^n f\left(\frac{k}{n}\right) - \frac{f(0)-f(1)}{2}| \leq |\frac{1}{4n}f'(0) - \frac{1}{4n}f'(1)| + \frac{\epsilon}{2}$$

Sending $n \to \infty$ and noticing that ϵ is arbitrary we get

$$\lim_{n\to\infty} |n\int_0^1 f(x)dx - \sum_{k=1}^n f\left(\frac{k}{n}\right) - \frac{f(0)-f(1)}{2}| = 0$$

and hence the limit evaluates to $\frac{f(0)+f(1)}{2}$. This finishes the proof. □

Problem 6.5. *Given continuous functions $\alpha : [0,1] \to \mathbb{R}$ and $\beta : [0,1] \to [0,1)$, define functions $f_n : [0,1] \to \mathbb{R}$ by the recursion*

$$f_{n+1}(x) = \alpha(x) + \int_0^x \beta(t) f_n(t)\ \mathrm{d}t$$

with $f_0 \equiv 0$. Prove that, for each $x \in [0,1]$, the limit

$$f(x) = \lim_{n\to\infty} f_n(x)$$

exists an compute its value.

Proof. Let $X = \mathcal{C}([0,1];\mathbb{R})$, which is a complete metric space when equipped with the distance induced by the supremum norm $||\cdot||$. Define $T : X \to X$ by

$$(Tf)(x) = \alpha(x) + \int_0^x \beta(t) f(t)\ dt.$$

If $f, g \in X$ then by compactness there exists a $\xi \in [0,1]$ such that

$$\sup_{x\in[0,1]} |(Tf)(x) - (Tg)(x)| = |(Tf - Tg)(\xi)|.$$

Then we get

$$\begin{aligned} ||Tf - Tg|| &= \left|\int_0^\xi \beta(t)(f(t) - g(t))\ dt\right| \\ &\leq \int_0^\xi |\beta(t)(f(t) - g(t))| dt \\ &\leq |\xi||\beta||||f - g|| < ||f - g||. \end{aligned}$$

Hence T is a contraction and has a fixed point by the Banach fixed point theorem. This proves the limit exists, since the fixed point is obtained by iterating, $f = \lim_{n\to\infty} \Gamma^{(n)} f_0$. □

Problem 6.6. *Let $f, g : \mathbb{R}^2 \to \mathbb{R}$ be continuously differentiable functions such that g attains value zero at least one point. Suppose that $\nabla g \neq 0$ everywhere and assume (x_0, y_0) is a point such that*

$$f(x_0, y_0) = \inf\{f(x,y) : x, y \in \mathbb{R}, g(x,y) = 0\}.$$

Show that then there is a $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0).$$

Proof. Set $S = g^{-1}(\{0\})$. (S is in fact a properly embedded submanifold). First recall that $\nabla g(x,y)$ is orthogonal the tangent space $T_{(x,y)}S$. To see this, suppose $\alpha : (-\epsilon, \epsilon) \to S$ is a local parametrization of S such that $\alpha(0) = p$, whose existence is guaranteed by the implicit function theorem. Then

$$0 = \frac{d}{dt}\Big|_{t=0} g(\alpha(t)) = \langle \nabla g(p), \alpha'(0) \rangle.$$

Moreover taking $p = (x_0, y_0)$, we see

$$0 = \frac{d}{dt}\Big|_{t=0} f(\alpha(t)) = \langle \nabla f(p), \alpha'(0) \rangle.$$

Since (x_0, y_0) is an extremum of $f(\alpha(t))$, it follows that $\nabla f(p)$ is also orthogonal to T_pS, and hence is parallel to $\nabla g(p)$, which is what we wanted to show. □

Problem 6.7. *A metric ρ in a metric space (X, ρ) is said to be an* ultrametric *if*

$$\forall\, x, y, z \in X : \qquad \rho(x,y) \le \max\{\rho(x,z), \rho(y,z)\}.$$

Prove that, in this metric, every open ball is closed and every closed ball is open.

Proof. Let $\overline{B}(x;r)$ be a closed ball of radius $r > 0$. We contend that for any $y \in \overline{B}(x;r)$ there exists $\delta > 0$ such that $B(y;\delta) \subset \overline{B}(x;r)$. If $\rho(x,y) < r$ the claim is trivial, so suppose $\rho(x,y) = r$. Assume for sake of contradiction that for all $\delta > 0$ there exists $z \in B(y;\delta)$ such that $\rho(z,y) > r$. Then assuming δ is chosen small enough, $\rho(y,z) \le \max\{\rho(y,x), \rho(z,x)\} < \delta$, a contradiction. This proves that the closed metric ball is open.
Similarly, we prove that $X \setminus B(x;r)$ is open. Let $y \in X \setminus B(x;r)$. It suffices to consider $\rho(y,x) = r$. Then as above assume for all $\delta > 0$ there is $z \in B(y;\delta)$ such that $\rho(z,x) < r$. Then $\rho(y,x) \le \max\{\rho(y,z), \rho(z,x)\} < r$ a contradiction. Therefore $B(x;r)$ is closed.

Remark 6.8. *An example of an ultrametric is the metric induced by the p-adic norm on $\mathbb{Q}$.*

□

Problem 6.9. *An orthogonal $n \times n$ matrix A is called elementary if the corresponding linear transformation $L_A : \mathbb{R}^n \to \mathbb{R}^n$ fixes an $(n-2)$-dimensional subspace. Prove that every orthogonal matrix M is a product of at most $(n-1)$ elementary orthogonal matrices.*

Proof. M is orthogonal if and only if there exists an orthogonal matrix A such that

$$AMA^{-1} = \operatorname{diag}\left(\lambda_1, \ldots, \lambda_k, \begin{pmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{pmatrix}, \ldots, \begin{pmatrix} \cos\theta_m & -\sin\theta_m \\ \sin\theta_m & \cos\theta_m \end{pmatrix}\right)$$

for eigenvalues $\lambda_i = \pm 1$, $\theta_i \in \mathbb{R}$, and $2m + k = n$. We can take the elementary matrices E_j to be the matrix with the same jth diagonal block and with the other blocks equal to the identity. Thus $AMA^{-1} = AE_1 \cdots E_{m+k}A^{-1}$. Notice $m + k < n$ except in the case $m = 0$. □

Problem 6.10. *Let $A = (a_{ij})$ be a 2×2 real matrix such that*

$$a_{11}^2 + a_{12}^2 + a_{21}^2 + a_{22}^2 < \frac{1}{10}.$$

Prove that $I + A$ is invertible.

Proof. To prove that $I + A$ is invertible, it suffices to show that $\det(I + A) \neq 0$.

$$\begin{aligned}\det(I + A) &= (1 + a_{11})(1 + a_{22}) - a_{12}a_{21} \\ &= 1 + a_{11} + a_{22} + a_{11}a_{22} - a_{12}a_{21} \\ &\geq 1 - 2\sqrt{\frac{1}{10}} - \frac{1}{10} > 0\end{aligned}$$

where the second last inequality is by the AM-GM inequality. This shows the result as desired. □

Problem 6.11. *Let $v_1 = (0, 1, x)$,$v_2 = (1, x, 1)$, and $v_3 = (x, 1, 0)$. Find all $x \in \mathbb{R}$ for which $\{v_1, v_2, v_3\}$ are linearly independent over $\mathbb{R}$. Similarly, find all $x \in \mathbb{R}$ for which $\{v_1, v_2, v_3\}$ are linearly independent over $\mathbb{Q}$.*

Proof. v_1, v_2, v_3 are linearly dependent iff

$$0 = \det \begin{pmatrix} 0 & 1 & x \\ 1 & x & 1 \\ x & 1 & 0 \end{pmatrix} = x(2 - x^2).$$

Over $\mathbb{Q}$ the only solution is $x = 0$. Over $\mathbb{R}$ the solutions are $x = 0, \pm\sqrt{2}$. □

Problem 6.12. *Let S be a subset of $Mat(3, \mathbb{C})$, the set of 3×3 matrices over $\mathbb{C}$. The set S is called dense if every matrix in $Mat(3, \mathbb{C})$ is a limit of a sequence of matrices in S.*

Prove that the set of matrices with distinct eigenvalues is dense in $Mat(3, \mathbb{C})$.

Prove that the set of matrices with one Jordan block is not dense in $Mat(3, \mathbb{C})$.

Proof.

1. Please refer to Problem 9, Spring 2015.
2. For convenience, we use S to denote the set of matrices with one Jordan block. Let $M \in S$, and we use C_M to denote the characteristic polynomial of M. Since M has only one Jordan block, C_M has a double root and thus the discriminant $\Delta C_M = 0$. Note that ΔC_M is an algebraic combination of the coefficient of C_M which turn out to be an algebraic combination of the entries of M, $f : M \mapsto C_M$ is continuous on $Mat(3, \mathbb{C})$. Given a matrix N with 3 distinct eigenvalues, $f(N) > 0$. Suppose there is $\{M_n\}_n \subset S$ such that $M_n \to N$, then $f(M_n) \to f(N)$ and thus $f(N) = 0$, which is impossible. Thus S cannot be dense, as desired. □

Problem 6.13. *Let*

$$M_n = \begin{pmatrix} a & b & 0 & \dots & 0 & 0 \\ -b & a & b & \dots & 0 & 0 \\ 0 & -b & a & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & a & b \\ 0 & 0 & 0 & \dots & -b & a \end{pmatrix}$$

where $a, b > 0$. Prove that $\det M_n > 0$ for all n. Prove that the limit

$$\lim_{n\to\infty} \det M_n$$

exists and compute its value.

Proof. Set $d_n = \det M_n$ for convenience. By expanding by minors we get the recurrence $d_n = ad_{n-1} + b^2 d_{n-2}$. $d_1 = a$ and we set $d_0 = 1$. The characteristic polynomial $x^2 - ax - b^2$ has roots $r_\pm = \frac{1}{2}(a \pm \sqrt{a^2+4b^2})$, so the solution is given by

$$d_n = cr_+^n + c'r_-^n$$

for some constants c, c'. $d_0 = 1$ gives $c + c' = 1$ and $d_1 = a$ gives $c - c' = \frac{a}{\sqrt{a^2+4b^2}}$. Solving, we get

$$c = \frac{a + \sqrt{a^2+4b^2}}{2\sqrt{a^2+4b^2}}, \quad c' = -\frac{a - \sqrt{a^2+4b^2}}{2\sqrt{a^2+4b^2}}.$$

Hence,

$$d_n = \frac{1}{\sqrt{a^2+4b^2}}\left(r_+^{n+1} - r_-^{n+1}\right).$$

However, $\log d_n$ need not converge as $n \to \infty$. □

Problem 6.14. *Let A be a real symmetric matrix, $n \geq 4$, and $v_1, ...v_4$ are non-zero vectors. Suppose $Av_k = (2k-1)v_k$ for all $1 \leq k \leq 4$. Prove that $v_1 + 2v_2 \perp 3v_3 + 4v_4$.*

Proof. Since A is real symmetric, by spectral theorem there is some orthogonal matrix P such that PAP^{-1} is diagonal. Since v_k for $1 \leq k \leq 4$ are eigenvectors with distinct eigenvalues, we may also assume that $Pv_k = a_k e_k$, where e_k is the standard Euclidean basis vectors and $a_k \in \mathbb{R}$. It follows that

$$\begin{aligned}\langle v_1 + 2v_2, 3v_3 + 4v_4\rangle &= \langle Pv_1 + 2Pv_2, 3Pv_3 + 4Pv_4\rangle \\ &= \langle a_1e_1 + 6a_2e_2, 15a_3e_3 + 28a_4e_4\rangle \\ &= 0\end{aligned}$$

where the first step is because P is orthogonal. Thus $v_1 + 2v_2 \perp 3v_3 + 4v_4$ as desired. □

§7 Fall 2017

Problem 7.1. *Let V be the vector space of polynomials with complex coefficients with degree at most 3.*

(a) Show that V is an inner product space with $\langle f, g\rangle = \int_{-1}^{1} f(t)\overline{g(t)}dt$.

(b) Find an orthonormal basis of V.

Proof. (a) Sesquilinearity follows from linearity of the integral. Also,

$$\langle f, f\rangle = \int_{-1}^{1} |f(t)|^2 dt \geq 0$$

with equality iff $f \equiv 0$. Hence it is a Hermitian inner product.

(b) Routine Gramm-Schmidt orthogonalization. Starting with the basis $\{1, x, x^2, x^3\}$ we get

$$1, x, x^2 - \frac{1}{3}, x^3 - \frac{3}{5}x.$$

Normalizing, we get

$$\frac{1}{2}, \frac{3}{2}x, \frac{5}{12}\left(x - \frac{1}{3}\right), \frac{175}{92}\left(x^3 - \frac{3}{5}x\right).$$

□

Problem 7.2. *Let $n \geq 1$ be an integer and A and B be $n \times n$ matrices.*

(a) Show that AB and BA have the same characteristic polynomial if A is invertible.

(b) Is the result true without assuming invertibility? Prove your claim.

Proof. (a) If A is invertible,

$$P_{AB} = \det(AB - \lambda I) = \det(A)\det(B - \lambda A^{-1}) = \det(B - \lambda A^{-1})\det(A) = \det(BA - \lambda I) = P_{BA}$$

as desired.

(b) The conclusion still holds. Since invertible matrices are dense, we choose $A_n \to A$ such that A_n are invertible matrices. Then $P_{A_n B} \to P_{AB}$ and $P_{BA_n} \to P_{BA}$ pointwise. But $P_{A_n B} = P_{B_n A}$ for all n, so $P_{AB} = P_{BA}$, as desired.

□

Problem 7.3. *Solve the system of linear differential equations*

$$\begin{cases} x_1'(t) &= 6x_1 - x_2 \\ x_2'(t) &= 2x_1 + 3x_2 \end{cases}.$$

Proof. Writing in vector form the system is $v'(t) = Av(t)$ where $A = \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}$ and $v = (x_1, x_2)$. It is routine to calculate the eigenvalues and eigenvectors, giving the general solution

$$v(t) = C_1 e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + C_2 e^{4t} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

for some constants C_1, C_2. □

Problem 7.4. *Let V be a vector space over $F = \mathbb{R}$ and let $V^* = \mathrm{Lin}_F(V, F)$ be the dual space. Let $\mathcal{B} = \{e_i\}_{i \in I}$ be a basis of V. For each $i \in I$, define the dual forms $e_i^\#$ by*

$$e_i^\#(e_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

(a) Show that the vectors $\mathcal{B}^* = \{e_i^\#\}_{i \in I}$ are linearly independent in V^*.

(b) Give necessary and sufficient conditions on V for these vectors to form a basis of V^*. Prove your claim.

Proof. (a) If $J \subset I$ is finite and $\sum_{j \in J} a_j e_j^\# = 0$, then evaluating the expression at e_j for each $j \in J$ shows $a_j = 0$, proving linear indpendence.

(b) If V is finite dimensional then for any $f \in V^*$ we can write

$$f = \sum_{i \in I} f(e_i) e_i^\#.$$

Hence B^* spans V^* and is therefore a basis.
Now suppose V is infinite dimensional. We claim $\dim V^* < \dim V$ (in cardinality). We know $V \cong (F^{\mathcal{B}})_0$, the set of functions from $\mathcal{B}$ to F with finite support. e.g. take the basis of

characteristic functions $\delta_b : \mathcal{B} \to F$, $\delta_b(b) = 1$, $\delta_b(v) = 0$ if $v \neq b$. Then $\dim(F^{\mathcal{B}})_0 = |\mathcal{B}| = \dim V$. We also know $V^* \cong F^{\mathcal{B}}$. Therefore it suffices to consider these to spaces.
Notice we have the countable subfield $\mathbb{Q} \subset F$. We claim that

$$\dim_F (F^{\mathcal{B}})_0 = \dim_{\mathbb{Q}} (\mathbb{Q}^{\mathcal{B}})_0 < \dim_{\mathbb{Q}} \mathbb{Q}^{\mathcal{B}} \leq \dim_F F^{\mathcal{B}}. \tag{3}$$

Observe that both $F_0^{\mathcal{B}}$ and $\mathbb{Q}_0^{\mathcal{B}}$ have the basis $\{\delta_b \mid b \in \mathcal{B}\}$, so the first equality (3) is manifest. For the middle inequality, for each finite subset $S \subset \mathcal{B}$ we set $\chi_S = \{f \in \mathbb{Q}_0^{\mathcal{B}} \mid f \text{ has finite support}\}$. We have

$$\mathbb{Q}_0^{\mathcal{B}} = \bigcup_{S \subset \mathcal{B}} \chi_S$$

where the union is over finite subsets of $\mathcal{B}$. Then $|\mathbb{Q}_0^{\mathcal{B}}| \leq |\mathcal{B}| \cdot |\aleph_0| = |\mathcal{B}|$. Since $|\mathbb{Q}_0^{\mathcal{B}}| \geq |\mathcal{B}|$ is obvious, we conclude $|\mathbb{Q}_0^{\mathcal{B}}| = |\mathcal{B}|$. On the other hand, for $S \subseteq \mathcal{B}$ not necessarily finite define $f_S : \mathcal{B} \to \mathbb{Q}$ by $f_S(b) = 1$ if $b \in S$ and $f_S(b) = 0$ otherwise. Then clearly $\{f_S \mid S \subseteq \mathcal{B}\}$ is a basis of $\mathbb{Q}^{\mathcal{B}}$, and therefore $|\mathbb{Q}^{\mathcal{B}}| = |P(\mathcal{B})|$, where $P(\mathcal{B})$ is the powerset of $\mathcal{B}$. By Cantor's diagonal argument, $|P(\mathcal{B})| > |\mathcal{B}|$ and the inquality is proved.

Finally, for the last inequality of (3) let $\{\alpha_k\}$ be a basis of F as a $\mathbb{Q}$-vector space and let $f_1, \ldots, f_n \in \mathbb{Q}^{\mathcal{B}}$ be linearly independent over $\mathbb{Q}$ such that

$$\sum_{i=1}^{n} r_i f_i = 0 \qquad \text{for some } r_1, \ldots, r_n \in \mathbb{R}.$$

Then writing $r_i = \sum_j q_j \alpha_{ij}$ for some $q_j \in \mathbb{Q}$ we see

$$\sum_{i,j} q_j \alpha_{ij} f_i = 0.$$

But since the f_i's are linearly independent over $\mathbb{Q}$, we must have $\sum_j q_j \alpha_{ij} = 0$ for each i. It follows that $q_j = 0$ for all j since the α_{ij}'s are linearly independent over $\mathbb{Q}$. Hence $r_i = 0$ for all $i = 1, \ldots, n$, proving that the f_i's are also independent over $\mathbb{R}$.

□

Problem 7.5. *Let V and W be two infinite dimensional vector spaces over $\mathbb{C}$. Let $\mathcal{L}(V, W)$ be the $\mathbb{C}$-vector space of $\mathbb{C}$-linear maps from V to W.*

(a) Is $X = \{f \in \mathcal{L}(V, W) \mid f \text{ has finite rank}\}$ a subspace of $\mathcal{L}(V, W)$?

(b) Same question for $Y = \{f \in \mathcal{L}(V, W) \mid \ker f \text{ is finite}\}$

(c) What is the intersection $X \cap Y$?

Proof. (a) Yes, since $\operatorname{Im}(f + g) \subseteq \operatorname{Im} f + \operatorname{Im} g$.

(b) No, for example id and $-$id have trivial kernel but their sum is the 0 operator, whose kernel is V

(c) $X \cap Y = \emptyset$ since a version of rank-nullity still holds for infinite dimensions:

$$\infty = \dim V = \ker f + \operatorname{rank} f.$$

□

Problem 7.6. *For each of the following three fields F, is it true that every symmetric matrix $A \in M_{2\times 2}$ is diagonalizable?*

(a) For $F = \mathbb{R}$.

(b) For $F = \mathbb{C}$.

(c) For $F = \mathbb{Z}/3\mathbb{Z}$.

Supply proofs/counterexamples (or cite relevant theorems) for all parts of this problem.

Proof. (a) Yes, real symmetric matrices are normal and hence diagonalizable.

(b) No, consider $A = \begin{pmatrix} -1 & i \\ i & 1 \end{pmatrix}$. $A^2 = 0$ and nilpotent matrices cannot be diagonalizable.

(c) No, consider $A = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$. The characteristic polynomial $x^2 - 2$ has no roots.

□

Problem 7.7. *Let $\{a_n\}_{n=1}^{\infty}$ be a non-increasing sequence of positive real numbers such that $\sum_{n=1}^{\infty} a_n < \infty$. Prove that*

$$\lim_{n\to\infty} na_n = 0$$

Proof. We will separate the cases of odd and even indices. Since $\sum a_n < \infty$, the partial sums S_n form a Cauchy sequence. Therefore, for any $\varepsilon > 0$

$$2na_{2n} \leq 2 \sum_{k=n+1}^{2n} a_k = 2(S_{2n} - S_n) < 2\varepsilon$$

for all n sufficiently large, as $\{a_n\}$ is non-increasing. This proves $\lim_{n\to\infty} 2na_{2n} = 0$. For odd indices, we have

$$0 \leq (2n+1)a_{2n+1} = \frac{2n+1}{2n} 2na_{2n+1} \leq \frac{2n+1}{2n} 2n \cdot a_{2n}.$$

Taking limits we conclude $\lim_{n\to\infty}(2n+1)a_{2n+1} = 0$ by the squeeze theorem. □

Alternate Proof. By Cauchy Condesation test,

$$\sum_{n=0}^{\infty} 2^n a_{2^n} \quad \text{converges.}$$

If $m \in \mathbb{N}$ we can choose n such that $2^n \leq m \leq 2^{n+1}$. Then

$$2^n a_{2^{n+1}} \leq m a_m \leq 2^{n+1} a_{2^n}.$$

We're done by the squeeze theorem. □

Problem 7.8. *Let $a < b$ be real numbers and $f : [a, b] \to \mathbb{R}$ a function such that $L(x) = \lim_{y\to x} f(y)$ exists for all $x \in [a, b]$ (with one-sided limits at $x = a, b$).*

1. *Prove that L is continuous on $[a, b]$.*
2. *Prove that $\{x \in [a, b] : f(x) \neq L(x)\}$ is countable.*

3. *Prove that f is Riemann integrable.*

Proof. 1. Notice that

$$\lim_{y\to x} L(y) = \lim_{y\to x}(\lim_{z\to y} f(z)) = \lim_{z\to x} f(z) = L(x)$$

showing that L is continuous as desired.

2. Define $E_\epsilon := \{x : |f(x) - L(x)| \geq \epsilon\}$, we want to show that E_ϵ is countable. Suppose $x \in E_\epsilon$, then for the given ϵ, there is an $\delta_x > 0$ such that as long as $|y - x| < \delta_x$, $|f(y) - L(x)| < \epsilon/2$, and $|L(y) - L(x)| < \epsilon/2$ and thus $|f(y) - L(y)| < \epsilon$, meaning that $y \notin E_\epsilon$. Thus points in E_ϵ are isolated and thus finite since $[a, b]$ is compact. Thus

$$\{x \in [a, b] : f(x) \neq L(x)\} = \bigcup_{n=1}^{\infty} E_{1/n}$$

showing that it is countable as a countable union of finite sets.

3. By part 2, f is discontinuous on a countable set, which has measure zero. By Lebesgue's criterion of integrability, f is Riemann integrable.

□

Problem 7.9. *Let $a < b$ be real numbers and $f : [a, b] \to \mathbb{R}$ a continuous function such that $\int_b^a f(x)x^n dx = 0$ for each integer $n \geq 0$. Prove that $f \equiv 0$.*

Proof. Let $\varepsilon > 0$. By Stone-Weierstrass theorem, there exists a polynomial $P(x)$ such that $\sup_{x\in[a,b]} |f(x) - P(x)| < \varepsilon$. It follows by linearity that

$$\int_a^b f(x)P(x) = 0.$$

Now note that

$$\begin{aligned} 0 \leq \int_a^b f(x)^2 dx &\leq \int_a^b f(x)(f(x) - P(x)) + f(x)P(x)dx \\ &\leq \varepsilon \int_a^b f(x)dx \end{aligned}$$

As ε was arbitrary, we must have $\int_a^b f(x)^2\ dx = 0$. It follows that $f(x) = 0$ for almost every $x \in [a, b]$. By continuity, f must vanish identically on $[a, b]$. □

Problem 7.10. *Let (X, ρ) be a complete metric space and $f : X \to X$ a function. Writing f^n for the n-th interate of f, denote*

$$c_n = \sup_{x\neq y} \frac{\rho(f^n(x), f^n(y))}{\rho(x, y)}.$$

Assuming that $\sum_{n\geq 1} c_n < \infty$, prove that f has a unique fixed point in X.

Proof. It's clear that $\rho(x, y) \leq c_n\rho(f^n(x), f^n(y))$ for all n. Choose an arbitrary $x_0 \in X$, and for $n > 0$ define recursively $x_n = f(x_{n-1})$. It follows that $\rho(x_{n+1}, x_n) \leq c_n\rho(x_1, x_0)$. We claim that $\{x_n\}$ is a Cauchy sequence in X. Let $\varepsilon > 0$. First note that the partial sums S_n of $\{c_n\}$ form a Cauchy

sequence (in $\mathbb{R}$), so there exists N large such that for all $m, n > N$ we have $|S_n - S_m| < \varepsilon/\rho(x_1, x_0)$. We obtain

$$\begin{aligned}\rho(x_n, x_m) &\leq \rho(x_n, x_{n-1}) + \rho(x_{n-1}, x_{n-2}) + \cdots + \rho(x_{m+1}, x_m) \\ &\leq \rho(x_1, x_0) \sum_{k=m}^{n} c_k < \varepsilon.\end{aligned}$$

Hence $\{x_n\}$ is Cauchy and converges to some $a \in X$ since X is complete. Obviously, $a = f(a)$ is the desired fixed point.
To prove uniqueness, suppose a' is another fixed point. $\rho(a, a') = \rho(f(a), f(a')) \geq c_1\rho(a, a')$ which implies $\rho(a, a') = 0$. □

Problem 7.11. *Prove Young's inequality: let $p, q \in (1, \infty)$ obey $\frac{1}{q} + \frac{1}{p} = 1$. Then for each $a, b \geq 0$,*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Proof. This is a special case of Holder's inequality. To begin, we prove a technical lemma:
Lemma. $a^\lambda b^{1-\lambda} \leq \lambda a + (1 - \lambda)b$.

To prove the lemma, we set $t = \frac{a}{b}$, and thus it is reduced to showing $t^\lambda \leq \lambda t + (1 - \lambda)$. By elementary calculus, $t^\lambda - \lambda t$ strictly increases for $t < 1$, and strictly decreases for $t > 1$. Hence the maximum is taken at $t = 1$, evaluated to be $1 - \lambda$. It follows that $t^\lambda - \lambda t \leq 1 - \lambda$, and combining the terms we obtain the result.

Now apply the lemma with $\lambda = \frac{1}{p}$ (and automatically $1 - \lambda = \frac{1}{q}$), $a = a^p$, and $b = b^q$ we obtain the desired result. □

Problem 7.12. *Let X be a compact metric space and $C(X)$ the space of continuous real-valued functions on X endowed with the supremum norm. Let $F \subset C(X)$ be non-empty. Prove the following version of Arzela-Ascoli's theorem:*

F is compact $\iff$ F is closed, bounded and equicontinuous.

Give precise definitions of all terms used in this equivalence.

Proof. Since we have already proved one version of Arzela-Ascoli (See Problem 10, 11 of Spring 2014), some slight modifications of the proofs will work for this one. □

§8 Spring 2017

Problem 8.1. *Let M be an $n \times n$ real matrix and with transpose M^T . Prove that M and $M \cdot M^T$ have the same range (image).*

Proof. First of all it's clear that $\mathrm{Im}(MM^T) \subset \mathrm{Im}(M)$. We claim that $\dim\ker(MM^T) = \dim\ker M^T$. Suppose the claim holds, it follows by rank-nullity that $\dim\mathrm{Im}(MM^T) = \dim\mathrm{Im}\, M^T = \dim\mathrm{Im}\, M$ and hence $\mathrm{Im}(MM^T) = \mathrm{Im}(M)$, showing the result. It remains to establish the claim.

Notice that $\ker M^T \subset \ker(MM^T)$. Conversely, suppose $x \in \ker(MM^T)$, $MM^Tx = 0$ and hence $0 = x^TMM^Tx = (M^Tx)^T(M^Tx) = \|M^Tx\|^2$, meaning that $M^Tx = 0$ and $x \in \ker M^T$. This finished the claim and concludes the proof. □

Problem 8.2. *Let $a, b, c, d \in \mathbb{R}$ and*

$$M = \begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & 0 & 0 \\ 0 & c & 3 & -2 \\ 0 & d & 2 & -1 \end{pmatrix}$$

1. *Determine conditions on a, b, c, d so that there is only one Jordan block for each eigenvalue of M in the Jordan form of M.*
2. *Find the Jordan form of M when $a = c = d = 2$ and $b = -2$.*

Proof. 1. We calculate the characteristic polynomial of M is $c_M(x) = (x-1)^4$. In order for there to be one Jordan block, we must have $\dim\ker(M - I) = 1$. Since

$$M - I = \begin{pmatrix} 0 & 0 & a & b \\ 0 & 0 & 0 & 0 \\ 0 & c & 2 & -2 \\ 0 & d & 2 & -2 \end{pmatrix}$$

we require $c \neq d$, $a \neq -b$.

2. In this case $\dim\ker(M - I) = 2$ and $\dim\ker(M - I)^2 = 4$, so the only possibility is two 2×2 blocks. The Jordan form is

$$M = S \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} S^{-1}.$$

□

Problem 8.3. *Let M be an $n \times n$ matrix. Suppose M is orthogonal and symmetric.*

1. *Prove that if M is positive definite then M is the identity.*
2. *Does the answer change if M is only positive semidefinite?*

Proof. 1. Since M is symmetric, by spectral theorem $M = SDS^{-1}$ for some D diagonal. M is positive definite, so the diagonal entries of D must be all positive real numbers, and since M is orthogonal, the diagonal entries of D, being eigenvalues of M, must all have absolute value 1. Hence $D = I$ and $M = I$, as desired.

2. The answer won't change if M is only positive semi-definite, since the above argument will work with the only modification being that the diagonal entries of D are non-negative real numbers.

□

Problem 8.4. *Let us define $F(x)$ as the following determinant:*

$$F(x) = \det \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & x & x^2 & x^3 & x^4 \\ x & 1 & x & x^2 & x^3 \\ x^2 & x & 1 & x & x^2 \\ x^3 & x^2 & x & 1 & x \end{pmatrix}.$$

Compute $F'(0)$.

Proof. Denote the matrix as $A(x)$. Jacobi's formula yields $F'(0) = F(0)\operatorname{tr}(A(0)^{-1}A'(0))$. We have

$$A(0) = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Then it is easy to calculate $\det A(0) = 1$ and

$$A(0)^{-1} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & -1 & -1 & -1 & -1 \end{pmatrix}, \qquad A'(0) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

Finally we get $\operatorname{tr}(A(0)^{-1}A'(0)) = -1$ so $F'(0) = -1$. □

Problem 8.5. *Suppose that V is a finite-dimensional vector space over the field $\mathbb{C}$, and $T : V \to V$ is a linear transformation. Further suppose that $F(x) \in \mathbb{C}[x]$ is a polynomial. Show that the linear transformation $F(T)$ is invertible if and only if $F(x)$ and the minimal polynomial of T have no common factors.*

Proof. Let the minimal polynomial of T be $m(x) = \prod(x - \lambda_i)^{\alpha_i}$ and let $F(x) = c\prod(x - z_j)^{\beta_j}$. Then $\det F(T) = c\prod \det(T - z_j)^{\beta_j}$. This equals zero iff $z_j = \lambda_i$ for some i and j i.e. when F and m have common factors. □

Problem 8.6.

1. *Let V be the vector space of real $n \times n$ matrices. Prove that the form $\langle A, B\rangle = \operatorname{trace}(A^T B)$ defines an inner product on V.*
2. *For $n = 2$, find an orthonormal basis in V.*

Proof.

1. We first check that $\langle \cdot, \cdot \rangle$ is bilinear. Notice that

$$\begin{aligned} \langle kA_1 + A_2, B\rangle &= \operatorname{trace}((kA_1 + A_2)^T B) = \operatorname{trace}(kA_1^T B + A_2^T B) \\ &= k\operatorname{trace}(A_1^T B) + \operatorname{trace}(A_2^T B) = k\langle A_1, B\rangle + \langle A_2, B\rangle \end{aligned}$$

and linearity of the other variable is verified similarly. To see that $\langle \cdot, \cdot \rangle$ is symmetric, we note that

$$\langle B, A\rangle = \operatorname{trace}(B^T A) = \operatorname{trace}(A^T B)^T = \operatorname{trace}(A^T B) = \langle A, B\rangle$$

Eventually, we need to show that $\langle \cdot, \cdot \rangle$ is positive definite. To see this, we first note that $A^T A$ is a positive semi-definite matrix since for every $v \in \mathbb{R}^n$

$$v^T A^T A v = (Av)^T (Av) \geq 0$$

and thus the eigenvalues of $A^T A$ are all non-negative. Moreover, since $A^T A$ is a real symmetric matrix, by spectral theorem there is some $n \times n$ matrix P such that $A^T A = PDP^{-1}$ for some diagonal matrix D. Since the eigenvalues of $A^T A$ are all non-negative, all entries of D are non-negative so that

$$\langle A, A\rangle = \operatorname{trace}(A^T A) = \operatorname{trace}(D) \geq 0$$

and $\langle A, A\rangle = 0$ iff $\operatorname{trace} D = 0$. Since D has non-negative entries, $\operatorname{trace} D = 0$ iff $D = 0$ iff $A^T A = 0$. Therefore, we have proven that $\langle \cdot, \cdot \rangle$ is positive definite and thus a bilinear form.

2. Observe that
$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\}$$
is such an orthonormal basis.

□

Problem 8.7. *Prove the existence and uniqueness of a non-negative continuous function* $f : [0,1] \to [0,1]$ *satsifying*
$$f(x) = 1 - \left(\int_0^x t f(t) \right)^2$$

Proof. Denote by V bounded continuous functions $[0,1] \to [0,1]$. It can be verified that V is a complete metric space under the uniform norm. Define an operator $T : V \to V$ by $T(f)(t) = 1 - \left(\int_0^x t f(t)\right)^2$. Then
$$\begin{aligned} |T(f)(x) - T(g)(x)| &= \left| \int_0^x t(f(t) - g(t))dt \right| \cdot \left| \int_0^x t(f(t) + g(t))dt \right| \\ &\leq \left(\|f - g\|_u \int_0^1 t dt \right) \left(\|f + g\|_u \int_0^1 t dt \right) \\ &\leq \frac{1}{2} \|f - g\|_u \end{aligned}$$
where the last inequality utilizes the fact that $\|f+g\|_u \leq 2$. Then by Contraction Mapping Theorem there is a unique fixed point f of the operator T, and f is the unique solution to this equation. $0 < [\int_0^x t f(t)]^2 < 1$, so f is non-negative. This finishes the proof. □

Problem 8.8. *Show that there is a constant* C *so that*
$$\left| \frac{f(0) + f(1)}{2} - \int_0^1 f(x)\, dx \right| \leq C \int_0^1 |f''(x)|\, dx$$
for every C^2 *function* $f : \mathbb{R} \to \mathbb{R}$.

Proof. Integrating by parts, we find
$$\begin{aligned} \int_0^1 f(x)dx &= \left(x - \frac{1}{2} \right) f(x) \Big|_0^1 - \int_0^1 \left(x - \frac{1}{2} \right) f'(x)dx \\ &= \frac{f(0) + f(1)}{2} - \int_0^1 \left(x - \frac{1}{2} \right) f'(x)dx. \end{aligned}$$
We can further integrate the second term by parts to get
$$\begin{aligned} \int_0^1 (x - \frac{1}{2}) f'(x)dx &= \left(\frac{x^2}{2} - \frac{x}{2} \right) f'(x) \Big|_0^1 - \int_0^1 \left(\frac{x^2}{2} - \frac{x}{2} \right) f''(x)dx \\ &= - \int_0^1 \left(\frac{x^2}{2} - \frac{x}{2} \right) f''(x)dx. \end{aligned}$$
Combining these observations, we obtain
$$\begin{aligned} \left| \frac{f(0) + f(1)}{2} - \int_0^1 f(x)\, dx \right| = \left| \int_0^1 \left(\frac{x^2}{2} - \frac{x}{2} \right) f''(x)dx \right| &\leq \int_0^1 \left| \frac{x^2}{2} - \frac{x}{2} \right| \cdot |f''(x)|dx \\ &\leq \frac{1}{8} \int_0^1 |f''(x)|\, dx. \end{aligned}$$

□

Problem 8.9. *Let (X, d) be a bouned metric space and let $C(X)$ denote the space of bounded continuous real-valued functions on X endowed with the supremum norm. Suppose $C(X)$ is separable.*

1. *Show that for every $\epsilon > 0$, there is a countable set $Z_\epsilon \subseteq X$ such that*

$$\forall\, x \in X, \quad \exists\, z \in Z_\epsilon, \quad \textit{such that } \forall\, y \in X, \quad |d(x,y) - d(z,y)| < \epsilon.$$

2. *Deduce that X is separable.*

Proof 1. 1. Since $C(X)$ is separable, there is a countable dense subset which we denote by $\mathcal{F}$. We also define $f_x(y) := d(x,y)$, which is clearly a continuous function. Let $\epsilon > 0$, we further define

$$\mathcal{B} := \{g \in \mathcal{F} \mid \|g - f_x\|_u < \epsilon/3 \text{ for some } x\}$$

and we enumerate the elements by $\{g_n\}_n$. Now, let x_n be an element in X such that $|g_n(x_n)| < \epsilon/3$, and we define $Z_\epsilon := \{x_n\}$. We claim that Z_ϵ has the desired properties. First of all we need to show that Z_ϵ is well-defined. Such x_n exists since

$$|g(x)| = |g(x) - f_x(x)| \leq \|g - f_x\|_u < \epsilon/3$$

We then show that Z_ϵ has the desired properties. Let $x \in X$. There is some g_i such that $\|g_i - f_x\|_u < \epsilon/3$. Then let $z = x_i$. For $y \in X$,

$$|d(x,y) - d(z,y)| = |f_x(y) - f_{x_i}(y)| \leq |f_x(y) - g_i(y)| + |g_i(y) - f_{x_i}(y)| \leq \|f_x - g_i\|_u + \|g_i - f_{x_i}\|_u < \epsilon$$

which shows the result as desired.

2. Take $Z = \bigcup_n Z_{1/n}$, we claim that Z is a countable dense subset of X. Let $x \in X$ and $\epsilon > 0$. Take n large enough such that $1/n < \epsilon$. Then there is $z \in Z_{1/n}$ such that for all $y \in X$, $|d(x,y) - d(z,y)| < \epsilon$. It follows that

$$d(x,z) = \sup_{y \in X} |d(x,y) - d(z,y)| < \epsilon$$

showing that Z is indeed dense in X. Z is clearly countable being a countable union of countable sets. Hence X is separable.

□

Proof 2. By problem 5 of Spring 2019, every metric space X can be isometrically embedded to $C(X)$. Then since $C(X)$ is a separable metric space, it is clear that X, viewed as a subspace of $C(X)$, is separable. Then taking a countable dense subset of X would solve the first half.

This proof is undesirable in three ways: first of all, it borrows the idea from another problem; second, the purpose of this problem is to deduce 2 from 1, but we do it backwards; third, proving that a subspace of a separable metric space is separable is actually a non-trivial exercise, since taking the intersection of the subspace with the countable dense subset won't work. □

Problem 8.10. *Show that for each $t \in \mathbb{R}$ fixed, the function*

$$F(y;t) = y^4 + ty^2 + t^2 y$$

defined for all $y \in \mathbb{R}$, achieves its global minimum at a single point $y_0(t)$.

Proof. Let $t \in \mathbb{R}$. Note that $F(y;t) \to +\infty$ as $|y| \to +\infty$, so the global minimum must be achieved at some critical points since $F(\cdot, t)$ is smooth. Now we split to cases.

If $t > 0$, $F''(y;t) = 12y^2 + 2t > 0$, and hence there can be at most one critical point which must be the global minimum.

If $t = 0$, $F(y;t) = y^4$, which clearly has global minimum 0 achieved at a single point.

If $t < 0$, this is the more interesting case. Fix t and set $f(y) := y^4 + ty^2$, $g(y) := t^2 y$, for which we have $f'(y) = 4y^3 + 2ty$, $f''(y) = 12y^2 + 2t$, $g'(y) = t^2$, $g''(y) = 0$. Note that $|f(y)| \to +\infty$ as $|y| \to \infty$ as $F(y;t)$ does, so the global minimum of f is also achieved at critical points. Then f has critical points $y_1 = -\sqrt{-\frac{t}{2}}$, $y_2 = 0$, $y_3 = \sqrt{-\frac{t}{2}}$. Plugging in we see that $f(y_1) = f(y_3) = -\frac{t^2}{4} < 0$ is the global minimum of f. Since g is monotone increasing, the global minimum of $F(\cdot, t)$ is achieved at $(-\infty, y_3]$. However, $F''(y_3;t) = 12y_3^2 + 2t = -4t > 0$, showing that there can be only one critical point in this interval and that single point must be where global minimum is achieved.

□

§9 Fall 2018

Problem 9.1. *Let $\{a_n\}_{n\geq 1}$ be a sequence of non-negative numbers such that*

$$\sum_{n\geq 1} a_n \qquad \textit{diverges.}$$

Show that

$$\sum_{n\geq 1} \frac{a_n}{2a_n + 1} \qquad \textit{diverges.}$$

Proof. We will prove the contrapositive. Suppose $\sum \frac{a_n}{2a_n+1}$ converges. Then for n sufficiently large, we have

$$\frac{a_n}{2a_n + 1} < \frac{1}{4}$$

or rearranging, $a_n < 1/2$. It follows that

$$\frac{a_n}{2} < \frac{a_n}{2a_n + 1}.$$

Hence, after discarding finitely many terms, we see $\sum a_n < \infty$. □

Problem 9.2. *Let A be a connected subset of $\mathbb{R}^n$ such that the complement of A is the union of two separated sets B and C, that is*

$$\mathbb{R}^n \setminus A = B \cup C \qquad \textit{with} \qquad \bar{B} \cap C = B \cap \bar{C} = \emptyset.$$

Show that $A \cup B$ is a connected subset of $\mathbb{R}^n$.

Proof. Suppose that $A \cup B$ is separated, so $A \cup B \subset D \cup E$ where D, E are open subsets of $\mathbb{R}^n$ with empty intersection. Since A is connected, it is contained entirely in D or E. WLOG $A \subset D$. Let $U = E \cap (A \cup B) = E \cap B$. It suffices to show that U is 'clopen,' which would imply either $U = \mathbb{R}^n$ or $U = \emptyset$. If $U = \mathbb{R}^n$, then $E = B = \mathbb{R}^n$, a contradiction. On the other hand, if $U = \emptyset$ then $B \subset D$ so $A \cup B$ is connected.

To show U is open, let $p \in U$ be arbitrary. $p \notin \overline{A}$ since $E \cap A = \emptyset$ and it is not posible for $p \in E \cap \overline{A}$ since E is open. Moreover, $p \notin \overline{C}$ since $B \cap \overline{C} = \emptyset$. Therefore $p \notin \overline{A \cup C} \subset \overline{A} \cup \overline{C}$, which

implies $p \in \mathbb{R}^n \setminus \overline{A \cup C}$. This in turn implies $p \in E \cap (\mathbb{R}^n \setminus \overline{A \cup C})$, which is an open set relative to U. Hence U is open.

To show U is closed, let $p \in \overline{U}$. Since $\overline{U} = \overline{E \cap B} \subset \overline{E} \cap \overline{B}$, we know $p \in \overline{B}$ and $p \notin C$ since $\overline{B} \cap C = \emptyset$. Thus $p \in A \cup B$. $p \notin A$, for otherwise $p \in D$, which would imply $D \cap \overline{U} = \emptyset$ and hence $D \cap E = \emptyset$, a contradiction. It follows that $x \in B$. Moreover, by the previous reasoning $x \notin D$, and $x \in E$ so $x \in U$. This proves $\overline{U} = U$ so U is closed. □

Problem 9.3. *Let $f : [0,1] \to \mathbb{R}$ and $g : [0,1] \to [0,1]$ be two Riemann integrable functions. Assume that*

$$|g(x) - g(y)| \geq \alpha |x - y| \quad \textit{for any } x, y \in [0,1]$$

and some fixed $\alpha \in (0,1)$. Show that $f \circ g$ is Riemann integrable.

Proof. For convenience, we define

$$E := \{x \in [0,1] : f \circ g \text{ is discontinuous at } x\}$$

$$F := \{x \in [0,1] : f \text{ is discontinuous at } x\}$$

$$G := \{x \in [0,1] : g \text{ is discontinuous at } x\}$$

Now, suppose $x \in E \backslash G$, then $f \circ g$ is discontinuous at x while f is g is continuous at x, so f is discontinuous at $g(x)$. Then $g(x) \in F$, and $x \in g^{-1}(F \cap g[0,1])$, showing that $E \backslash G \subset g^{-1}(F \cap g[0,1])$. By our assumption

$$|g(x) - g(y)| \geq \alpha |x - y| \quad \text{for any } x, y \in [0,1]$$

the inverse $g^{-1} : g[0,1] \to [0,1]$ is α-Lipschitz. Since f is integrable, F has measure zero and so is $F \cap g[0,1]$. Since Lipschitz functions send null sets to null sets, $g^{-1}(F \cap g[0,1])$ also has measure zero and thus $E \backslash G$ has measure zero. Since g is integrable, G has measure zero, and thus E has measure zero. Thus $f \circ g$ is Riemann integrable, as desired. □

Problem 9.4. *Let $f : [0,1] \to \mathbb{R}$ be continuous on $[a,b]$ and differentiable on the open interval $(0,1)$. Assume that $f(0) = 0$ and f' is a decreasing function on $(0,1)$. Show that*

$$g(x) = \frac{f(x)}{x}$$

is a decreasing function on $(0,1)$.

Proof. First note that

$$g'(x) = \frac{xf'(x) - f(x)}{x^2}$$

so it suffices to show $xf'(x) - f(x) < 0$ for all $x \in (0,1)$. By the mean value theorem, for each $x \in (0,1)$ there exists $\xi \in (0,x)$ such that $f'(\xi)(x - 0) = f(x) - f(0)$, i.e. $f'(\xi) = f(x)/x$. Since f is decreasing and $\xi < x$, we know $f'(x) < f'(\xi)$. Hence $f'(x) < f(x)/x$, as desired. □

Problem 9.5. *Let $B = \{x \in \mathbb{R}^n | \, |x| \leq 1\}$ and let $g : \partial B \to \mathbb{R}$ be a 1-Lipschitz function.*

(a) Show that the function $f : B \to \mathbb{R}$ given by

$$f(x) = \inf_{y \in \partial B} (g(y) + |x - y|)$$

is 1-Lipschitz.

(b) *Show that the set* $M(g) = \{h : B \to \mathbb{R} |\ h$ *is 1-Lipschitz and* $h\big|_{\partial B} = g\}$ *is compact in the space of continuous functions on* B *endowed with the supremum norm.*

Proof. 1. First of all, for $x_1, x_2 \in B$,

$$f(x_1) - f(x_2) = \inf_{y \in \partial B}(g(y) + |x_1 - y|) - \inf_{y \in \partial B}(g(y) + |x_2 - y|)$$

Since $g(y) + |x_2 - y|$ is continuous of y and ∂B is compact, there is some $y_2 \in \partial B$ such that

$$g(y_2) + |x_2 - y_2| = \inf_{y \in \partial B}(g(y) + |x_2 - y|)$$

Then

$$f(x_1) - f(x_2) \leq g(y_2) + |x_1 - y_2| - g(y_2) - |x_2 - y_2| \leq |x_1 - x_2|$$

and similarly

$$f(x_2) - f(x_1) \leq |x_1 - x_2|$$

so that

$$|f(x_1) - f(x_2)| \leq |x_1 - x_2|$$

showing that f is 1-Lipschitz.

2. First of all, we notice that $M(g)$ is bounded, so every sequence in $M(g)$ contains a convergent subsequence. To show that $M(g)$ is compact, we can just use the standard limiting argument to verify that the limit of the convergence subsequence is in $M(g)$.

□

Problem 9.6. *For* $x \in (0, \infty)$ *let*

$$F(x) = \int_0^\infty \frac{1 - e^{-tx}}{t^{3/2}} dt.$$

Show that $F : (0, \infty) \to (0, \infty)$ *is well-defined, bijective, and of class* C^1*, and that its inverse is of class* C^1*.*

Proof. We first show that F is well-defined. To see this, we notice that

$$F(x) = \int_0^{1/2} \frac{1 - e^{-tx}}{t^{3/2}} dt + \int_{1/2}^\infty \frac{1 - e^{-tx}}{t^{3/2}} dt$$

It is clear that $F(x) > 0$. Moreover,

$$\int_{1/2}^\infty \frac{1 - e^{-tx}}{t^{3/2}} dt \leq \int_{1/2}^\infty \frac{dt}{t^{3/2}} < \infty$$

and

$$\int_0^{1/2} \frac{1 - e^{-tx}}{t^{3/2}} dt = \int_0^{1/2} t^{-1/2} x + O(t^{1/2} x^2) dt < \infty$$

So $F(x) \in (0, \infty)$.

Next, we notice by change of variables that

$$F(x) = x^{3/2} \int_0^{1/2} \frac{1 - e^{-tx}}{(tx)^{3/2}} dt = x^{1/2} \int_0^\infty \frac{1 - e^{-tx}}{t^{3/2}} dt = F(1) x^{1/2}$$

where $F(1) \in (0, \infty)$ by our previous argument. Then it is immediate that F is bijective, C^1, and has a C^1 inverse. This finishes the proof. □

Problem 9.7. *Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map with the property that*

$$T(T(x)) = T(x) \quad \text{for all } x \in \mathbb{R}^n.$$

Show that there exists $1 \le m \le n$ and a basis fo $\mathbb{R}^n$ such that, with respect to this basis, the entries of T satisty

$$T_{ij} = \begin{cases} 1 & \text{if } i = j \text{ and } 1 \le i \le m \\ 0 & \text{otherwise.} \end{cases}$$

Proof. We can write $\mathbb{R}^n = \ker T \oplus \operatorname{Im} T$. Let $\{e_1, \dots e_m\}$ be an ordered basis of $\operatorname{Im} T$ and extend it to a basis $\{e_1, \dots e_n\}$ of $\mathbb{R}^n$ so that $T(e_i) = 0$ for $m < i \le n$. Since $T|_{\operatorname{Im} T}$ has a trivial kernel, it follows that $\{T(e_1), \dots, T(e_m)\}$ is also a basis of $\operatorname{Im} T$. The matrix of T with respect to these bases is $\mathrm{Id}_m \oplus \mathbf{O}_{n-m}$ as desired. □

Problem 9.8. *Let X be a real $n \times n$ symmetric matrix and $z \in \mathbb{C}$ with $\operatorname{Im} z > 0$. Define $G = (X - z)^{-1}$. Show that*

$$\sum_{1 \le j \le n} |G_{ij}|^2 = \frac{\operatorname{Im} G_{ii}}{\operatorname{Im} z}. \tag{4}$$

Proof. Write $X = SDS^{-1}$ where $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ where $\lambda_i \in \mathbb{R}$. It follows that

$$G = S(D - z)^{-1} S^{-1}.$$

Since $(D - z)^{-1} = \operatorname{diag}((\lambda_1 - z)^{-1}, \dots, (\lambda_n - z)^{-1})$, the left hand side of (4) is

$$\begin{aligned} \sum_{1 \le j \le n} |S_{ij}(D - z)^{-1} S_{ij}^{-1}|^2 &= \sum_{1 \le j \le n} |S_{ij}(\lambda_i - z)^{-1} S_{ij}^{-1}|^2 \\ &= \sum_{1 \le j \le n} |\delta_{ij}(\lambda_i - z)^{-1}|^2 \\ &= |(\lambda_i - z)^{-1}|^2. \end{aligned}$$

Moreover, since S is a real matrix

$$\operatorname{Im} G_{ii} = \operatorname{Im} \frac{1}{\lambda_i - z} = \operatorname{Im} \frac{\lambda_i - \bar{z}}{|\lambda_i - z|^2} = \frac{\operatorname{Im} z}{|\lambda_i - z|^2}.$$

This proves the identity. □

Problem 9.9. *Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be linearly independent of the vector space (over $\mathbb{R}$) of linear mappings from $\mathbb{R}^n$ to $\mathbb{R}$. Show that for any $v \in \mathbb{R}^n$, there exists v_1 and v_2 such that*

$$v = v_1 + v_2, \quad f(v) = f(v_1) \quad \text{and} \quad g(v) = g(v_2).$$

Proof. Write $V = \mathbb{R}^n$. It sufficies to show that $\ker f + \ker g = V$. We know

$$\dim(\ker f + \ker g) = \dim \ker f + \dim \ker g - \dim(\ker f \cap \ker g). \tag{5}$$

Denote $W = \ker f \cap \ker g$. Since $\dim \ker g = \dim \ker g = n - 1$, it suffices to show $\dim W = n - 2$. Clearly we cannot have $\dim W < n - 2$, since then (5) is $> n$. Nor is is possible $\dim W = n$ (for then $f = g \equiv 0$). So suppose $\dim W = n - 1$. We can choose a complementary subspace for W so that $V = W \oplus \langle u \rangle$ for some vector $u \notin W$. But then $-g(u)f + f(u)f \equiv 0$, contradicting linear independence of f and g. □

Problem 9.10. *Let*

$$A = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

Calculate $\lim_{n\to\infty} A^n$.

Proof. We can diagonalize $A = BDB^{-1}$ where B is invertible and $D = \operatorname{diag}(1, \frac{1}{2}, \frac{1}{3})$. Note that $A^n = BD^nB^{-1}$ so

$$\begin{aligned} \lim_{n\to\infty} A^n &= B \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^{-n} & 0 \\ 0 & 0 & 3^{-n} \end{pmatrix} B^{-1} \\ &= B \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} B^{-1}. \end{aligned}$$

Calculating the eigenvector corresponding to 1, we get

$$\lim_{n\to\infty} A^n = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

□

Problem 9.11. *Let V be the space of all 3×3 real matrices that are skew-symmetric, i.e. $A^t = -A$. Prove that the expression*

$$\langle A, B\rangle = \frac{1}{2}\operatorname{Tr}(AB^t)$$

defines an inner product on V. Exhibit an orthonormal basis of V with respect to this inner product; rigorously justify your answer.

Proof. V is in fact $\mathfrak{o}(n)$, the Lie algebra of the group of orthogonal $n\times n$ matrices. An orthonormal basis is

$$e_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, e_2 \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

□

Problem 9.12. *Let V be a finite dimensional vector space. Let $T : V \to V$ be a linear transformation such that $T(W) \subseteq W$ for every subspace W of B with $\dim W = \dim V - 1$. Prove that T is a scalar multiple of the identity.*

Proof. Let $v \in V$, then the subspace generated by v is given by

$$\langle v\rangle = \bigcap_{W : v\in W} W$$

where the intersection is over subspaces W with $\dim W = \dim V - 1$. It follows that $T\langle v\rangle \subseteq \langle v\rangle$ since $T(W) \subseteq W$ for any such W in the intersection. Hence $T(v) = \lambda v$ for some scalar λ.
Now choose $w \neq v$. $T(w) = \lambda' w$ and $T(v + w) = \lambda''(v + w)$ for some scalars λ', λ''. But

$$\lambda''(v + w) = T(v + w) = T(v) + T(w) = \lambda v + \lambda' w$$

so we must have $\lambda = \lambda'$. Since v and w were arbitrary, we conlude $T = \lambda \cdot \mathrm{id}$. □

§10 Spring 2018

Problem 10.1. *Prove that the functions $e^t, e^{2t}, ..., e^{nt}$ are linearly independent in the sapce of continuous functions on the interval $[1, 2]$.*

Proof. For convenience, we denote $f_k(t) = e^{kt}$. Suppose

$$a_1 f_1 + ... + a_n f_n \equiv 0 \tag{6}$$

on $[1, 2]$, we want to show that $a_1 = ... = a_n = 0$. To see this, repeatedly taking derivatives of the above equation (where on endpoints we take one-sided derivatives) we get

$$\begin{aligned} a_1 f_1 + ... + n a_n f_n &\equiv 0 \\ a_1 f_1 + ... + n^2 a_n f_n &\equiv 0 \\ &\cdots \\ a_1 f_1 + ... + n^{n-1} a_n f_n &\equiv 0 \end{aligned}$$

which means that for any $t \in [1, 2]$ we have

$$\begin{pmatrix} 1 & \dots & 1 \\ 1 & \dots & n \\ \vdots & & \vdots \\ 1 & \dots & n^{n-1} \end{pmatrix} \begin{pmatrix} a_1 f_1(t) \\ a_2 f_2(t) \\ \vdots \\ a_n f_n \end{pmatrix} = 0$$

we know that the matrix on the left hand side is invertible, so we must have

$$a_1 f_1(t) = a_2 f_2(t) = ... = a_n f_n(t) = 0$$

for $t \in [1, 2]$. Since $f_k > 0$ on $[1, 2]$, we get (6) as desired. □

Problem 10.2. *Let A and B be two 5×5 matrices, such that $A^2 = A$, $B^2 = B$, and $1 - (A + B)$ is invertible. Prove that* $\operatorname{rank}(A) = \operatorname{rank}(B)$.

Proof. Since $1 - (A + B)$ is invertible, we have

$$\begin{aligned} \operatorname{rank} A &= \operatorname{rank}[(1 - A - B)A] = \operatorname{rank}(A - A^2 - BA) = \operatorname{rank}(BA) \\ &= \operatorname{rank}(B - B^2 - BA) = \operatorname{rank}[B(1 - A - B)] = \operatorname{rank} B \end{aligned}$$

as desired. □

Problem 10.3. *Let $A = (a_{ij})$ be a complex $n \times n$ matrix. Recall that matrix*

$$e^A = 1 + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

is well-defined. Suppose $e^A = 1 + A + A^2$. Prove or disprove: A is a zero matrix.

Proof. We claim that A is not necessarily a zero matrix. To see this, consider

$$A := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Then direct computation gives

$$A^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 0$$

Then

$$e^A = 1 + A = 1 + A + A^2$$

showing the claim as desired. If we want to construct a counterexample for each n, we simply consider an $n \times n$ matrix where the above 2×2 matrix is situated at the top-left corner. □

Problem 10.4. *Consider the following matrices:*

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \quad E = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad F = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Which pairs of these matrices are similar over $\mathbb{R}$? You must fully justify your answer.

Proof. To solve this problem, it suffices to determine whether they have the same Jordan canonical form, and since all matrices above have only one eigenvalue 1, it suffices to determine the number of Jordan blocks of each matrix. We know that given a matrix M with eigenvalue λ, the number of Jordan blocks corresponding to λ is $\dim \ker(M - \lambda I)$, so applying this result to the matrices above we see that A, B, C, D, E are pairwise similar, but F is not similar to any other matrix above. □

Problem 10.5. *Let A, B be two positive definite 2×2 matrices. Prove or disprove:*

1. *$A + B$ is also positive definite.*
2. *$AB + BA$ is also positive definite.*

Proof.

1. This conclusion is true. First of all it is clear that $A + B$ is symmetric as a sum of two symmetric matrices. Since A and B are both positive definite, for any $v \in \mathbb{R}^2$ we have

$$v^T(A + B)v = v^T Av + v^T Bv \geq 0$$

Morever, since $v^T Av, v^T Bv \geq 0$, $v^T(A + B)v = 0$ implies $v^T Av = v^T Bv = 0$, and since A, B are positive definite this means that $v = 0$. Thus $A + B$ is positive definite.

2. This conclusion is false. For example, let

$$A := \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B := \begin{pmatrix} 6 & 0 \\ 0 & 1 \end{pmatrix}$$

It is clear that A, B are symmetric, and it can be verified that the eigenvalues of A are $\frac{3 \pm \sqrt{5}}{2}$ and the eigenvalues of B are 6 and 1, so A and B are both positive definite. Meanwhile, direct computation shows that

$$AB = \begin{pmatrix} 24 & 7 \\ 7 & 2 \end{pmatrix}$$

whose determinant is -1 and cannot be positive definite.

□

Problem 10.6. *Compute the determinant of the following matrix:*

$$\begin{pmatrix} 1 & 2 & 4 & 8 & 16 \\ 2 & 4 & 8 & 16 & 1 \\ 4 & 8 & 16 & 1 & 2 \\ 8 & 16 & 1 & 2 & 4 \\ 16 & 1 & 2 & 4 & 8 \end{pmatrix}$$

Solution. Doing some row reductions and bruteforcing, we get the result 923521. This is not enlighening at all and hopefully the reader can come up with a better solution. □

§11 Spring 2019

Problem 11.1. *Let $C([0,1])$ denote the space of continuous real-valued functions on $[0,1]$ equipped with the distance*

$$d(f,g) = \sup_{x\in[0,1]} |f(x) - g(x)|.$$

Let X be the subset of $C([0,1])$ defined via

$$X = \{f : [0,1] \to \mathbb{R} : f(0) = 0 \text{ and } |f(x) - f(y)| \le |x - y|\}.$$

Show that X is connected and complete.

Proof. This is a rather easy exercise, so I choose to not fill in the details. To show connectedness, we show in particular X is convex. Completeness is a standard argument; given a Cauchy sequence in X we prove that its pointwise limit is its limit in X. □

Problem 11.2. *Let $a, b \in \mathbb{R}$ be two real numbers and consider a sequence $(a_n)_{n\in\mathbb{N}}$ defined recursively by*

$$a_0 = a, \quad a_1 = b, \quad a_n = \frac{a_{n-1} + a_{n-2}}{2} \quad \text{for } n \ge 2.$$

Prove that $\lim_{n\to\infty} a_n$ exists and compute its value.

Proof. We solve the recurrence explicitly. The characteristic polynomial is $2x^2 - x - 1 = (2x + 1)(x - 1)$ so

$$a_n = c \cdot 1^n + c' \left(\frac{-1}{2}\right)^n$$

for some constants c, c'. From $a_0 = c + c' = a$ and $a_1 = c - c'2 = b$ we get $c = \frac{a+2b}{3}$ and $c' = \frac{2}{3}(a - b)$. Hence

$$\lim_{n\to\infty} a_n = \frac{a + 2b}{3}.$$

□

Problem 11.3. *Let $a, b \in \mathbb{R}$ with $a \le b$ and let $f : [a,b] \to \mathbb{R}$ be a Riemann integrable function. Assume that there is a $\delta > 0$ such that $f(x) \ge \delta$ for all $x \in [a,b]$. Show that the function $\frac{1}{f}$ is Riemann integrable.*

Proof. Let $\epsilon > 0$ and choose a partition $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$ of $[a, b]$ such that $U(f, P) - L(f, P) < \epsilon/\delta^2$, where $U(f, P)$ and $L(f, P)$ are the upper and lower Riemann sums of f with respect to P, respectively. Set $M_i = \sup_{x\in[x_{i-1},x_i]} f(x)$ and $m_i = \inf_{x\in[x_{i-1},x_i]} f(x)$. Then

$$\begin{aligned}\sup_{x\in[x_{i-1},x_i]} \frac{1}{f(x)} - \inf_{x\in[x_{i-1},x_i]} \frac{1}{f(x)} &= \frac{1}{m_i} - \frac{1}{M_i} = \frac{M_i - m_i}{M_i m_i} \\ &\leq \frac{1}{\delta^2}(M_i - m_i).\end{aligned}$$

□

Thus, $U(1/f, P) - L(1/f, P) < \epsilon$ so $1/f$ is Riemann integrable.

Problem 11.4. *Let $a, b \in \mathbb{R}$ with $a < b$ and let $f, g : [a, b] \to \mathbb{R}$ be continuous functions that are differentiable on (a, b). Show that there exists some $\xi \in (a, b)$ such that $(f(b) - f(a))g'(\xi) = (g(b) - g(a))f'(\xi)$.*

Proof. We will take the regular version of Mean Value Theorem for granted, or more specifically Rolle's Theorem.
If $g(b) = g(a)$ then the result follows from Rolle's theorem, so assume $g(a) \neq g(b)$. Define $h(x) = f(x) - \alpha g(x)$, where $\alpha = \dfrac{f(b) - f(a)}{g(b) - g(a)}$. Then $h(a) = h(b)$. By Rolle's Theorem there exists $\xi \in (a, b)$ such that $h'(\xi) = 0$, i.e.

$$f'(\xi) = \frac{f(b) - f(a)}{g(b) - g(a)} \cdot g'(\xi).$$

□

Problem 11.5. *Show that each metric space can be embeded isometrically into a Banach space. Concretely, if (D, d) is a metric space, then there exists a Banach space $(X, |\cdot|)$ and a map $S : D \to X$ with $|S(x) - S(x')| = d(x, x')$ for $x, x' \in D$.*

Proof. Let $X = \{f : D \to \mathbb{R} : \ f \text{ is continuous } \}$ and let $|\cdot|$ be the supremum norm. We take for granted that $(X, |\cdot|)$ is a Banach space.
Choose any $x_0 \in D$ and define $S(x)$ by $S(x)(y) = d(x, y) - d(x_0, y)$. We calculate

$$\begin{aligned}\sup_{z\in D} |S(x)(z) - S(x')(z)| &= \sup_{z\in D} |d(x, z) - d(x_0, z) - (d(x', z) - d(x_0, x'))| \\ &= \sup_{z\in D} |d(x, z) - d(x', z)| \\ &= d(x, x').\end{aligned}$$

The last equality follows from the fact that $d(x, z) \leq d(x, x') + d(x', z)$. □

Problem 11.6. *For $n \geq 1$ let $f_n : \mathbb{R}_+ \to \mathbb{R}$, $f_n(x) = \frac{x}{n^2} e^{-x/n}$. Show that the sequence (f_n) converges to 0 unifomrly over $\mathbb{R}_+$, but $\lim_{n\to\infty} \int_0^\infty f_n(x)dx = 1$.*

Proof. Using L'Hospitals rule, we can see that $\frac{x}{n} e^{-x/n} \to 0$ as $x \to \infty$ and thus $|f_n| \leq \frac{M}{n}$ uniformly in $x > 0$, where in particular M is independent of n. Therefore we immediately see that $f_n \to 0$ uniformly on $\mathbb{R}_+$. Meanwhile, integrating by parts we get

$$\begin{aligned}\int_0^\infty \frac{x}{n^2} e^{-x/n}\, dx &= \frac{1}{n^2}\left(- nxe^{-x/n}\Big|_0^\infty + \int_0^\infty ne^{-x/n} dx\right) \\ &= \frac{1}{n}\int_0^\infty e^{-x/n} dx \\ &= 1\end{aligned}$$

as desired. This finishes the proof. $\square$

Problem 11.7. *For*

$$A = \begin{pmatrix} 1 & 0 & 3 \\ 3 & -5 & 3 \\ 1 & -1 & 2 \end{pmatrix}$$

write A^{-1} as a polynomial in A with real coefficients.

Proof. The characteristic polynomial of A is $-x^3 - 2x^2 + 13x - 1$. By Cayley-Hamilton theorem A satisfies $-A^3 - 2A^2 + 13 \cdot A - \mathrm{id} = 0$, i.e.

$$A(-A^2 - 2A + 13 \cdot \mathrm{id}) = \mathrm{id}.$$

We conclude $-A^2 - 2A + 13\,\mathrm{id} = A^{-1}$. $\square$

Problem 11.8. *Let V be the vector space of all 2×2 matrices with real entries. Let W be the subspace generated by all matrices of the form*

$$AB - BA$$

where $A, B \in V$. What is the dimension of W? Justify your answer.

Proof. We first note that

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

So $\dim W \geq 3$. Moreover, since $\mathrm{trace}(AB) = \mathrm{trace}(BA)$,

$$\mathrm{trace}(AB - BA) = 0$$

Therefore $1 \notin W$ and thus $\dim W < \dim V = 4$. It follows that $\dim W = 3$, as desired. $\square$

Problem 11.9. *Let V be the vector space over $\mathbb{C}$ of all complex polynomials of degree at most 10. Let $D : V \to V$ be the differentiation operator, so $Df(x) = f'(x)$. Find all eigenvalues and eigenvectors of e^D on V.*

Proof. Expanding we get

$$e^D f(x) = 1 + f'(x) + \frac{f''(x)}{2!} + \dots.$$

This is the Taylor expansion of $f(x+1)$ centered at x so $e^D f(x) = f(x+1)$. Therefore we are seeking polynomials such that $f(x) = \lambda f(x+1)$ for some $\lambda \in \mathbb{C}$. Clearly constants are eigenvectors with eigenvalue 1. We claim there are no eigenvectors with $n = \deg f \geq 1$. Let $\alpha_1, \dots, \alpha_n in \mathbb{C}$ be the zeros of f, not necessarily distinct. Then $\prod_{0<i\leq n}(x - \alpha_i)$ divides $f(x) - f(x+1)$. But $f(x) - f(x+1)$ has degree at most $n-1$ and so cannot have n roots, a contradiction. This proves the claim.

Remark 11.10. *Here the Taylor series terminates after* $\deg f$ *terms, but* $e^D f(x) = f(x+1)$ *is true for analytic functions as well. In that case* $f(x) = e^{x \log \lambda}$ *has eigenvalue* λ *for any* $\lambda \in \mathbb{R}$.

□

Problem 11.11. *Let* A *be an* $n \times n$ *complex diagonalizable matrix and* I *the* $n \times n$ *identity matrix. Show that the* $2n \times 2n$ *matrix*

$$B = \begin{pmatrix} A & I \\ 0 & A \end{pmatrix}$$

is not diagonalizable.

Proof. Notice that for any nonconstant polynomial $P(x)$ we have $P(B) = \begin{pmatrix} P(A) & P'(A) \\ 0 & P(A) \end{pmatrix}$. This can be easily verified by induction on basis polynomials x^k.

Let $m_B(x)$ and $m_A(x)$ be the minimal polynomials of B and A, respectively. We use the follow critierion: B is diagonalizable iff m_B is the product of distinct linear factors. Since $m_B(A) = 0$, it follows that $m_A | m_B$. Write $m_B(x) = m_A(x)P(x)$ for some polynomial P so $m_B'(x) = m_A'(x)P(x) + m_A(x)P'(x)$. Since $m_B'(A) = m_A'(x)P(A) = 0$, we must have $m_A | m_A' \cdot P$. Hence $\gcd(m_B, m_B') \neq 1$ which implies that m_B is not a product of distinct linear factors. □

Problem 11.12. *Let* A *be an* $n \times n$ *complex matrix. Prove that rank*$(A) =$ *rank*(A^2) *if and only if* $\lim_{\lambda \to 0} (A + \lambda I)^{-1} A$ *exists, where* I *is the identity matrix.*

Proof. First suppose the limit exists. If $v \in \ker A^2$ then the Neumman series reduces to

$$(A + \lambda I)^{-1} A v = \lambda^{-1} A \left(I + (-\lambda^{-1} A) + (-\lambda^{-1} A)^2 + \dots \right) v = \lambda^{-1} A v.$$

Since the limit as $\lambda \to 0$ exists we must have $Av = 0$. Hence $\ker A = \ker A^2$. It follows that $\operatorname{Im} A^2 = \operatorname{Im} A$.

Conversely, suppose $\operatorname{rank} A^2 = \operatorname{rank} A$. We claim $\mathbb{C}^n = \ker A \oplus \operatorname{Im} A$, in which case it suffices to check the limit exists on the two subspaces. First, $\ker A \cap \operatorname{Im} A = \{0\}$ since $\operatorname{Im} A^2 = \operatorname{Im} A$. The sum is the whole space by the rank-nullity theorem, so the claim is proved.
For $v \in \ker A$ clearly $(A + \lambda I)Av = 0$. A restricted to $\operatorname{Im} A$ is non-singular,

$$\lim_{\lambda \to 0} (A + \lambda I)^{-1} A v = v$$

for $v \in \operatorname{Im} A$. Hence the limit exists on $\mathbb{C}^n = \ker A \cap \operatorname{Im} A$. □

Problem 11.13. *Let* $A = (a_{ij})$ *be an* $n \times n$ *real matrix whose diagonal entries satisfy* $a_{ii} \geq 1$ *for each* $i = 1, \dots n$. *Suppose also that*

$$\sum_{i \neq j} a_{ij}^2 < 1.$$

Prove that A^{-1} *exists.*

Proof. Let $\|A\|$ denote the Frobenius norm of A, and let $B = \operatorname{diag}(a_{11}, \dots, a_{nn})$.. Assume that $Av = 0$ so $(A - B)v = -Bv$. Using the hypotheses, together with the fact that $|Av| \leq \|A\| \cdot |v|$,

$$|v|^2 \leq |Bv|^2 = |(A - B)v| \leq \|A - B\| \cdot |v| < |v|.$$

This implies $v = 0$, so $\ker A$ is trivial and therefore A is invertible. □